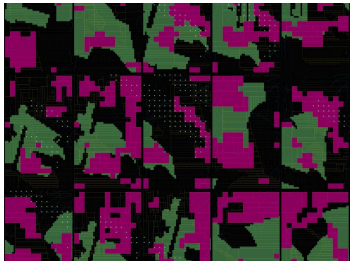
PORTFOLIO · 2023–2026

JOHN NORTHRUP

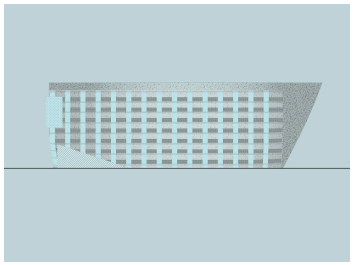
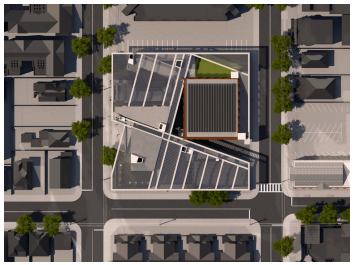
Designer & Digital Fabrication Specialist | Computational Workflows & Physical Systems

I am a designer and technical fabricator specializing in computational modeling, digital workflows, and physical construction. Grounded in hands-on material fabrication—from fine joinery to structural welding—and technical spatial design, my work bridges complex digital systems with buildable, physical space. I focus on technical design development, project management, and translating complex digital models into precise physical outcomes.

JCNORTHRUP@GMAIL.COM JCNORTHRUP.COM





CODE & AI

01	memoryMachine	SU26, M.ARCH THESIS	03
02	AMENITY DECK ENGINES	2026, SELF-DIRECTED COMPUTATIONAL STUDY	06
03	PROTOTYPE	SP26, PROTOTYPE, 3GB STUDIO	14

ARCHITECTURE

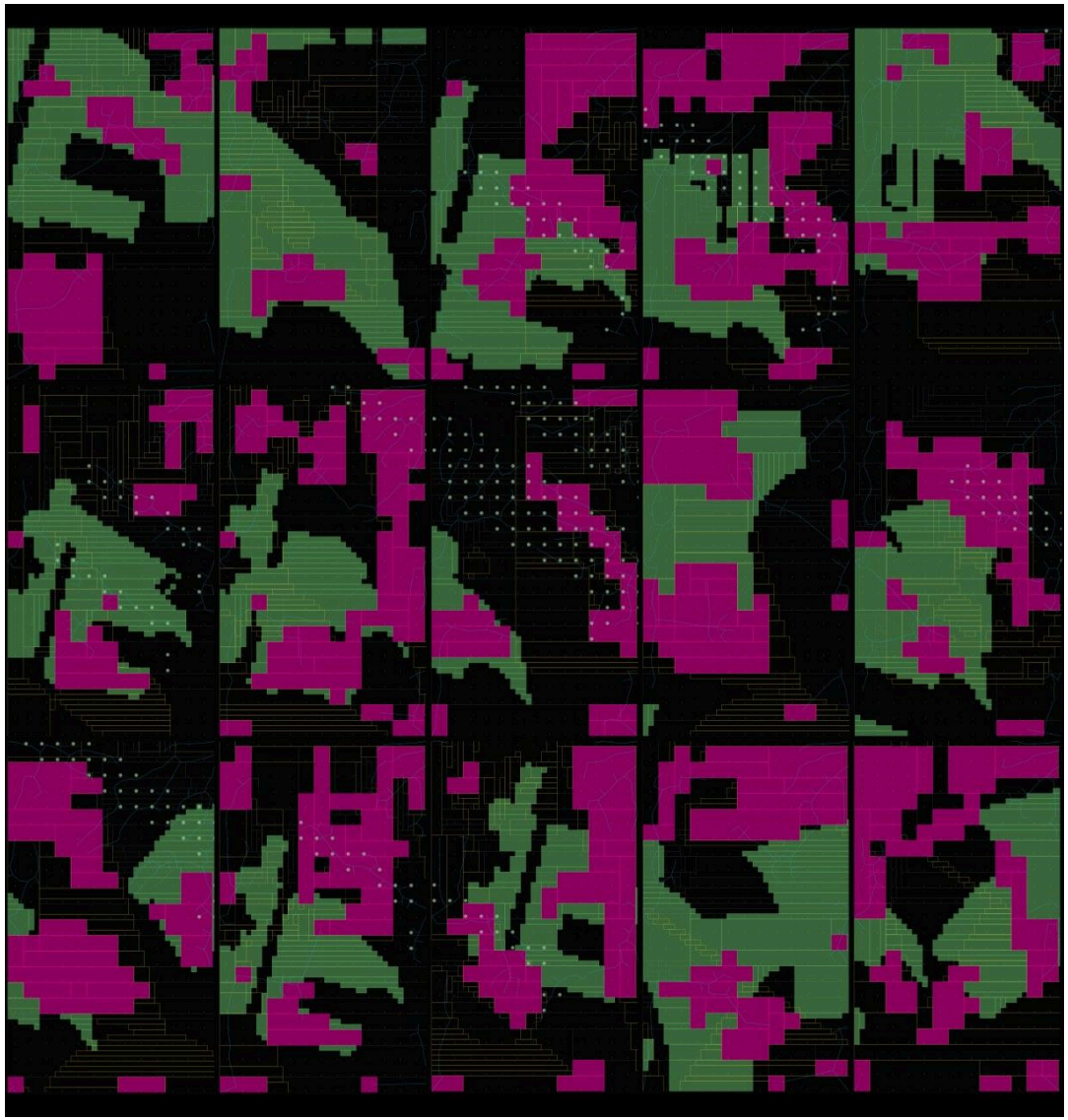
04	DESIGN SCHOOL IN MEXICO CITY	SP25, 2GB STUDIO, WITH NATE SMITH	17
05	LITTLE TOKYO THEATER	FA23, DS 1100, 1GA STUDIO	20
06	LA DANCE SHED	FA24, 2GA STUDIO, WITH MILAN SLEDGE	27

DRAWING & DOCUMENTATION

07	DESIGN DOCUMENTS	DESIGN DOCUMENTS	31
08	ADVANCED PROJECT DELIVERY	ADVANCED PROJECT DELIVERY	34

FABRICATION & MECHANICAL

09	RESILIENT FUTURES	2025, SCI-ARC RESILIENT FUTURES TASK FORCE	36
10	DETAILS, DETAILS	F25, DETAILS, DETAILS	39
11	ELF-O-MATIC-3000	FA22, MONSTER CITY STUDIOS	41



01

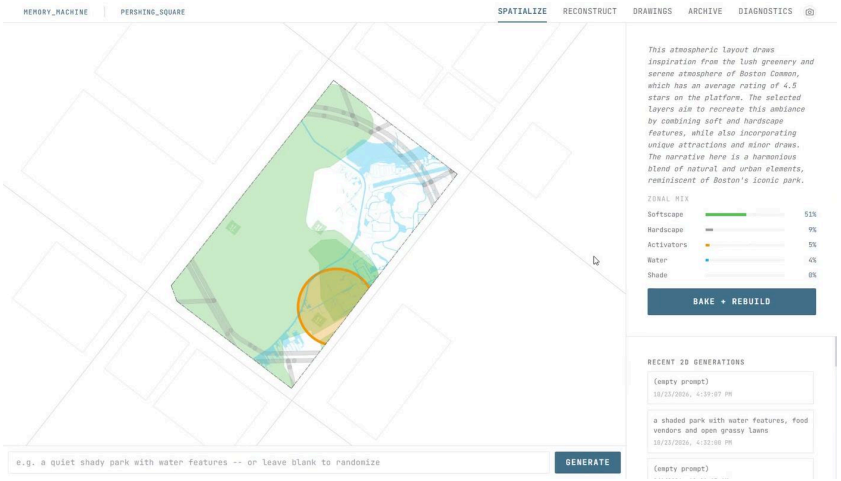
memoryMachine

SU26, M.ARCH THESIS
ADVISOR · CASEY REHM

memoryMachine is a computational design framework that transforms civic sentiment and archival memory into operative spatial data to revitalize under-utilized public space. Using Pershing Square in Downtown Los Angeles as a test bed—a civic site defined by a history of top-down, disconnected redesigns—the project proposes an adaptive spatial infrastructure in place of the fixed, one-time master plan.

At its core, a multimodal data ingestion engine aggregates web reviews, archival records, and spatial metrics, pairing qualitative sentiment with physical variables such as microclimate, canopy coverage, and circulation flows. To ensure equitable representation, the engine applies demographic weighting to mitigate digital access bias. These inputs are translated into spatial density fields and programmatic adjacencies, generating interventions—such as terraced landforms, water systems, and subterranean transit connections—as evidence-based spatial responses rather than predetermined formal gestures.

By merging expert analysis with experiential civic data, memoryMachine repositions the architect from sole author to system mediator. Public input shifts from a one-off consultation phase into a permanent, operative design parameter. The result is a self-regulating spatial framework that ingests usage patterns to continuously re-evaluate the site and adapt through phased physical interventions—establishing a democratic model in which public space evolves alongside the community it serves.



SOFTWARE | GENERATE A PLAN, REBUILD IT AS MASSING, DRAW IT (SPED UP) (VIDEO: JCNOR...



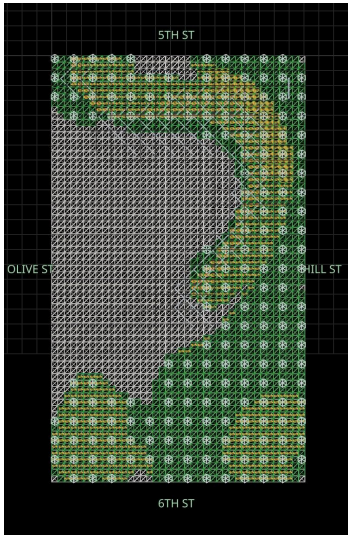
INSTALLATION | TIMBER SITE MODEL IN THE EXHIBITION HALL



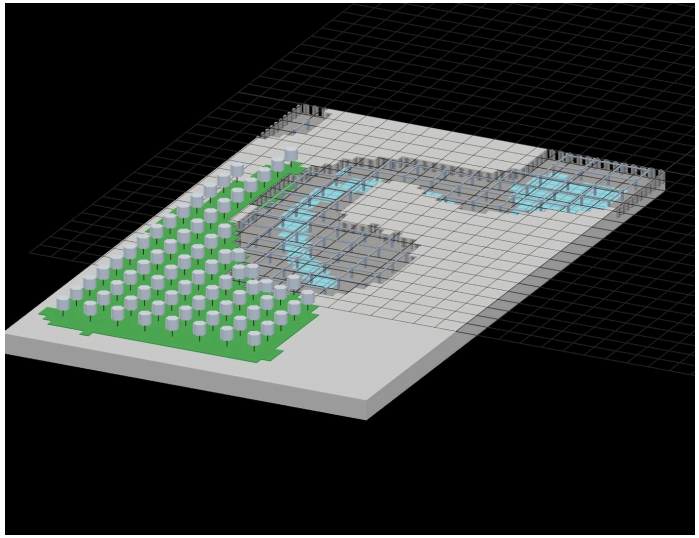
INSTALLATION | TIMBER SITE MODEL



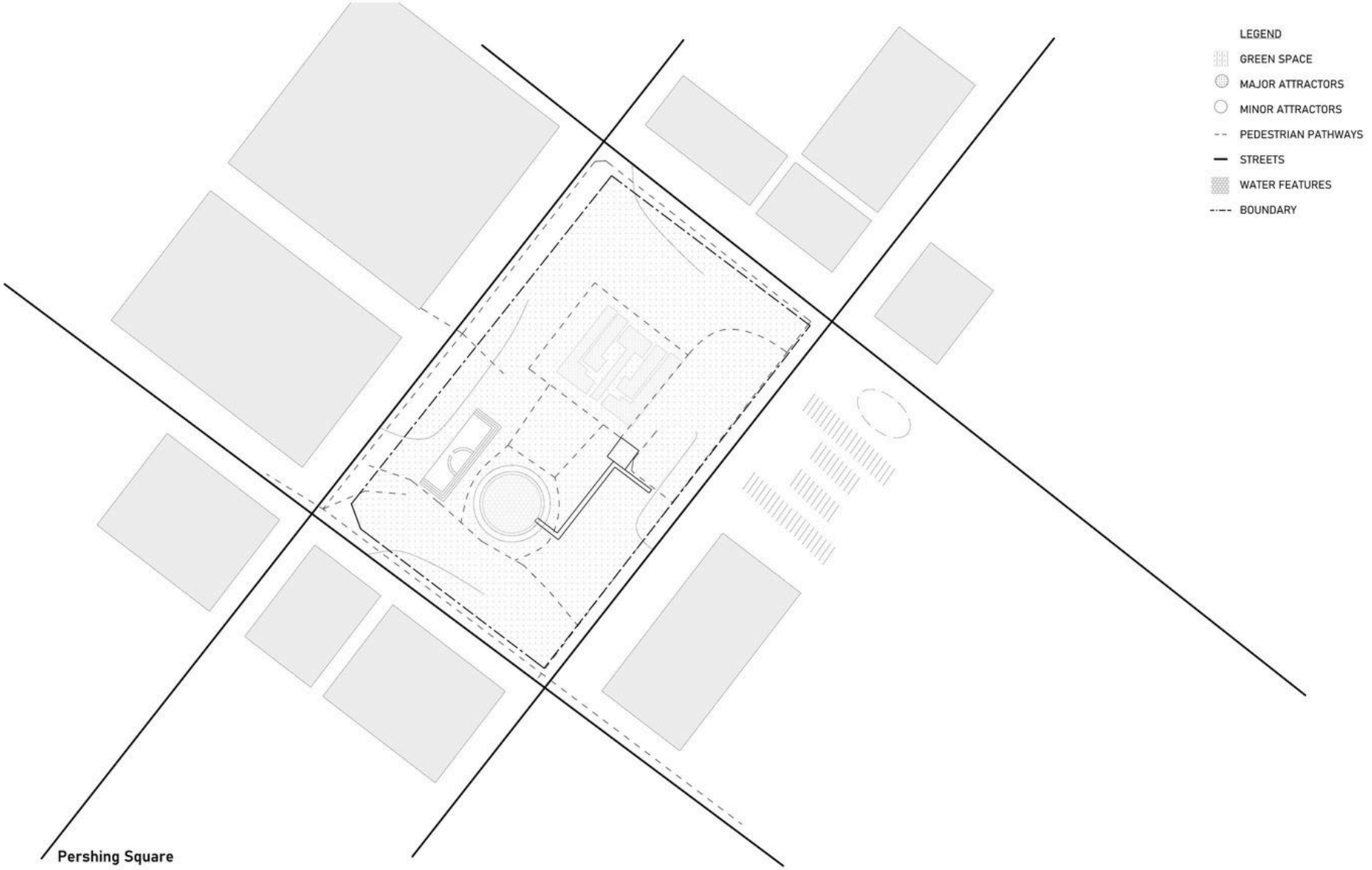
RENDER | THE GENERATED PARK IN ITS DOWNTOWN CONTEXT



GENERATED PLAN | PERSHING SQUARE



GENERATED MODEL | THE SAME SCHEME AS A MASSING



DRAWING | PERSHING SQUARE, LOS ANGELES: THE PROJECT SITE READ IN THE SYSTEM'S OWN TERMS



02

AMENITY DECK ENGINES

2026, SELF-DIRECTED COMPUTATIONAL STUDY
ROLE · DESIGN & SCRIPTING

A parametric workflow for a high-rise podium deck, balancing complex site constraints—mechanical clearances, egress cores, and hardscape—with three distinct privacy zones.

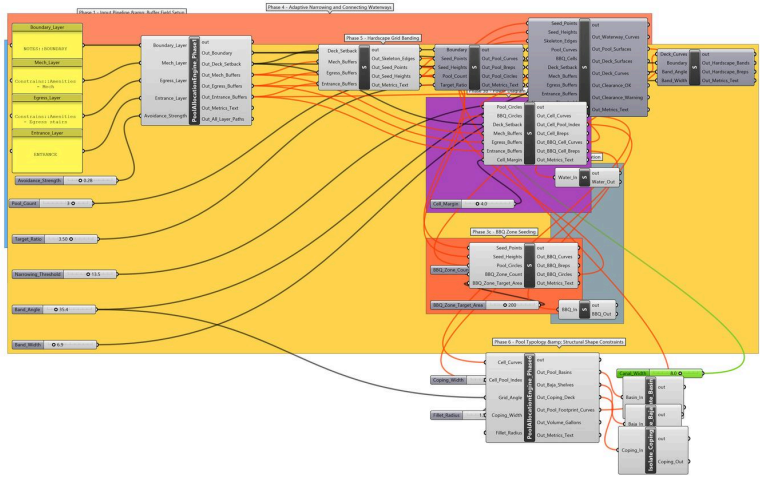
The initial engine deployed a six-phase pipeline: extracting Rhino boundary geometry, buffering fixed obstacles, and using a straight-skeleton algorithm to seed pool locations based on target deck-to-water ratios. A weighted Voronoi power diagram then resolved spatial overlaps into obstacle-clipped cells to automatically generate grid-aligned basins.

While mathematically solved, the Voronoi cells proved spatially arbitrary. Instead of forcing the output, the power diagram was repurposed as an analytical overlay, revealing exactly where programmatic density naturally wanted to settle.

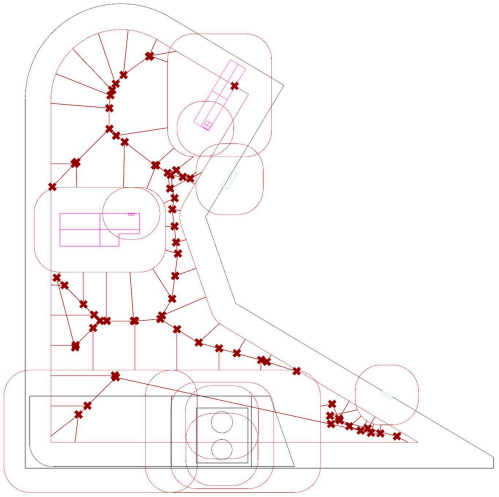
Using hand-tuned zonal curves derived from this initial analysis, a streamlined secondary definition handles the downstream geometric execution—building site-aligned lap basins, stepped coping, and integrated soaking pools directly from spatial intent. The algorithm evolved from an automated author into an executive instrument.

*The two photoreal perspectives are Rhino renders finished with AI enhancement. Every drawing and diagram here is output from the definitions.

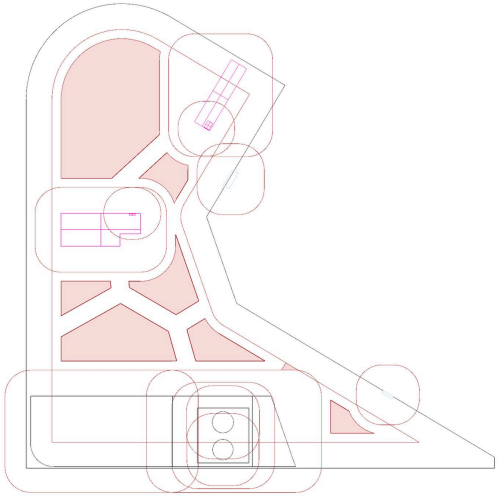
THE INITIAL ENGINE



DEFINITION | SIX-PHASE PIPELINE, GRASSHOPPER AND PYTHON

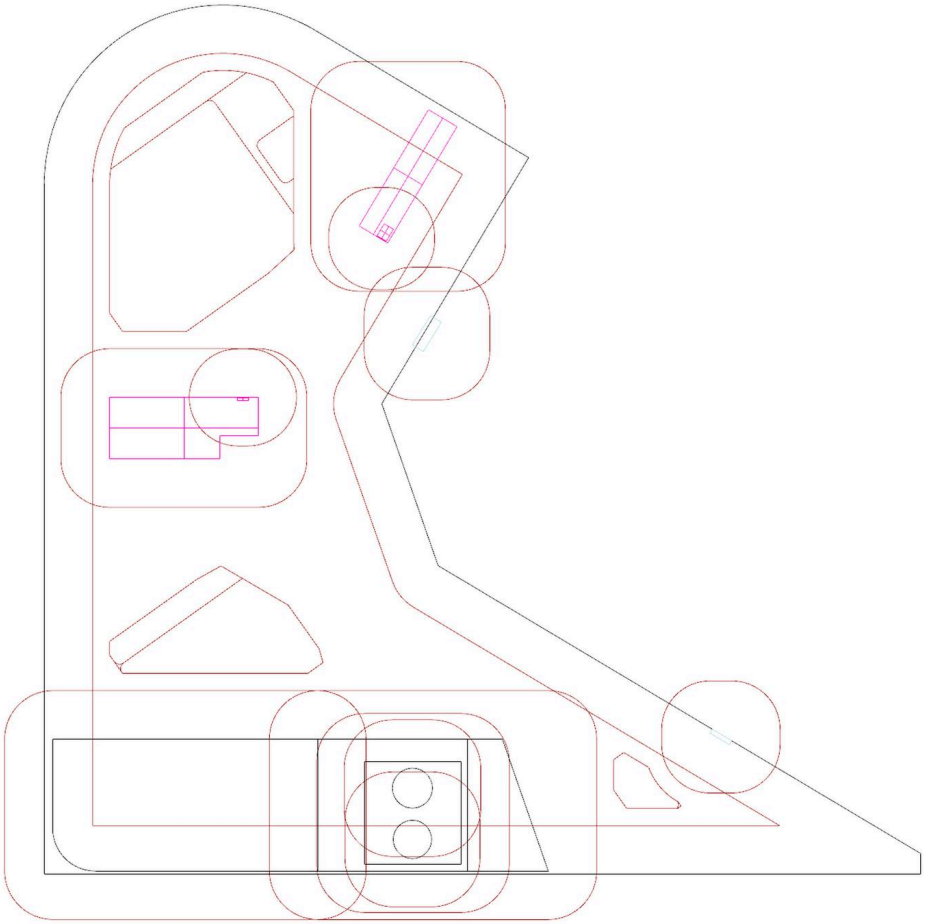


PHASE 2 | STRAIGHT SKELETON AND SEED POINTS

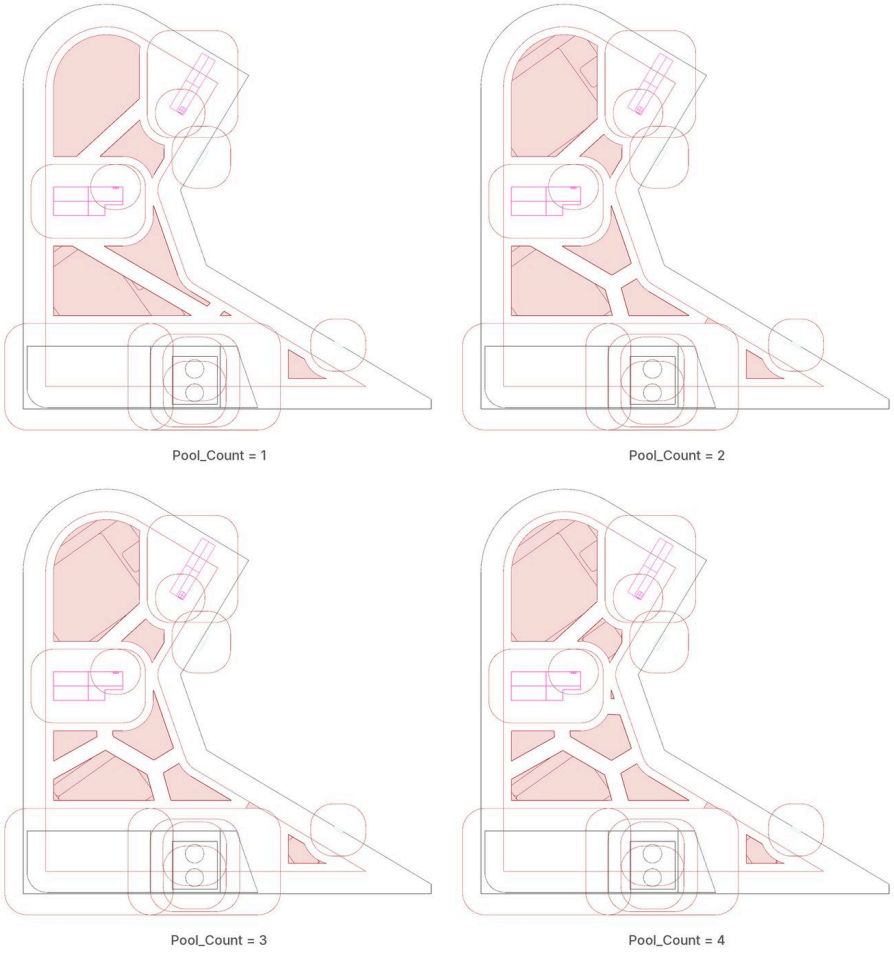


PHASE 3B | WEIGHTED VORONOI POWER DIAGRAM

THE INITIAL ENGINE

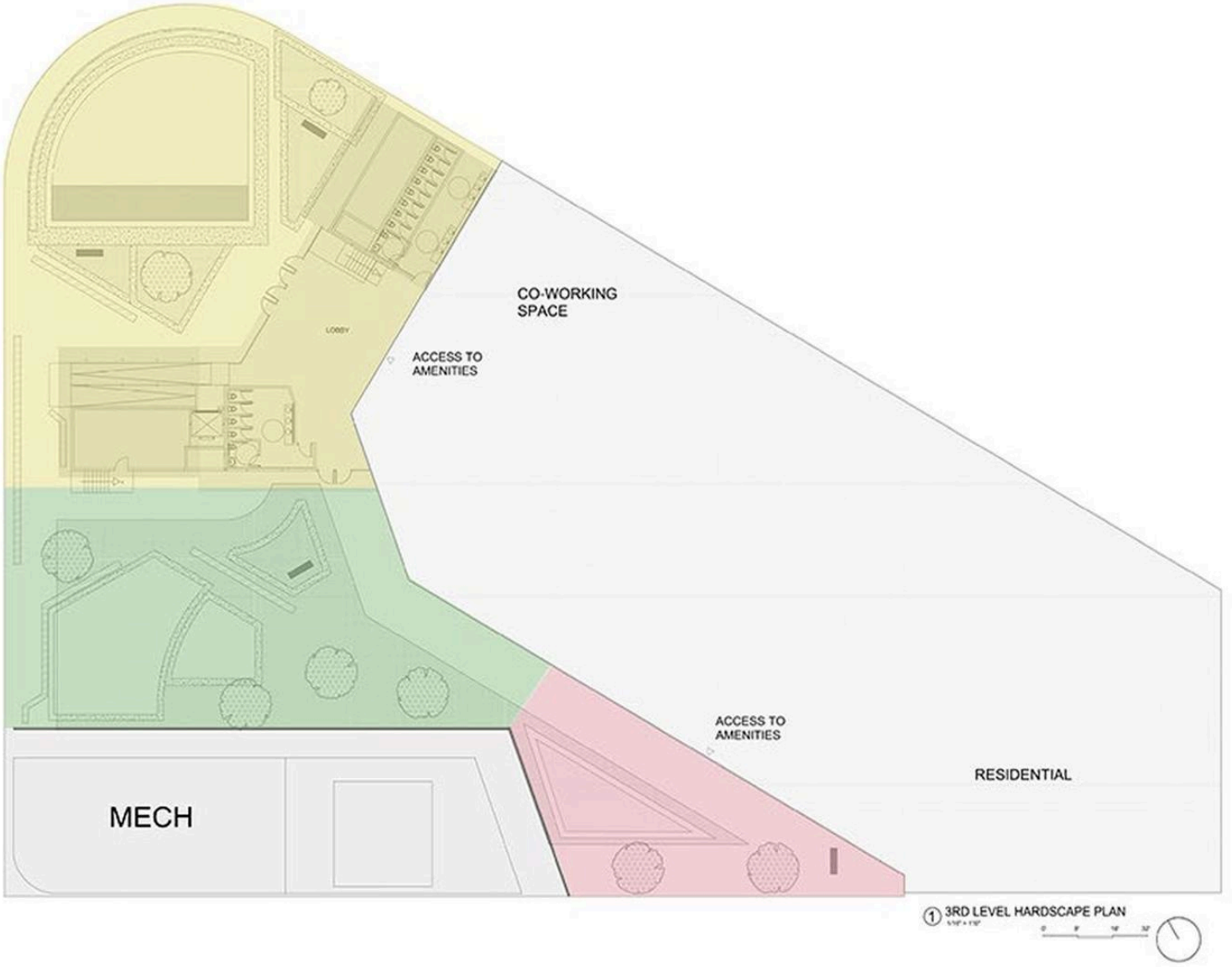


PHASE 6 | CELLS CONVERTED TO BUILDABLE BASINS



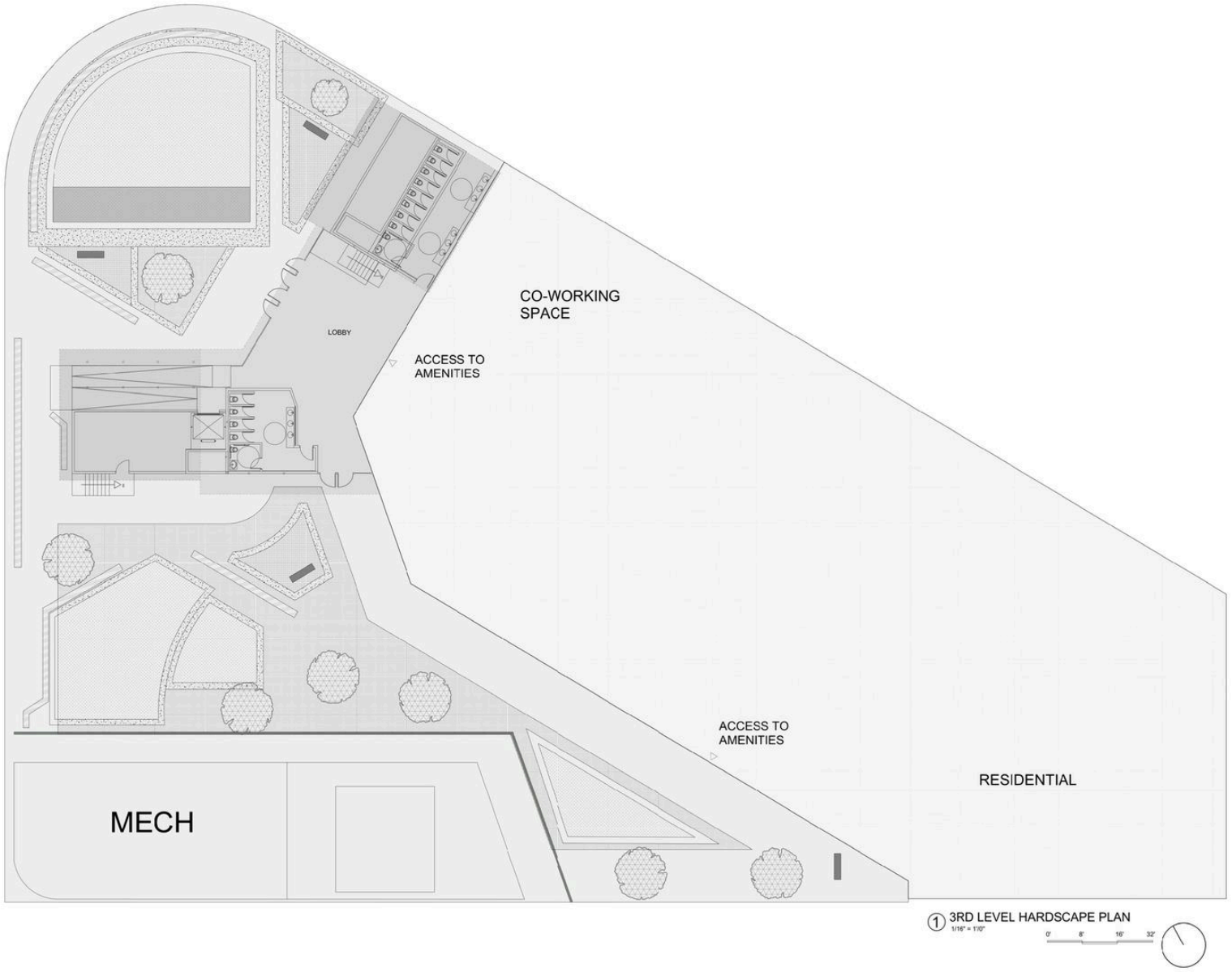
OPTION ARRAY | ONE DEFINITION, FOUR POOL COUNTS

THE SECONDARY DEFINITION



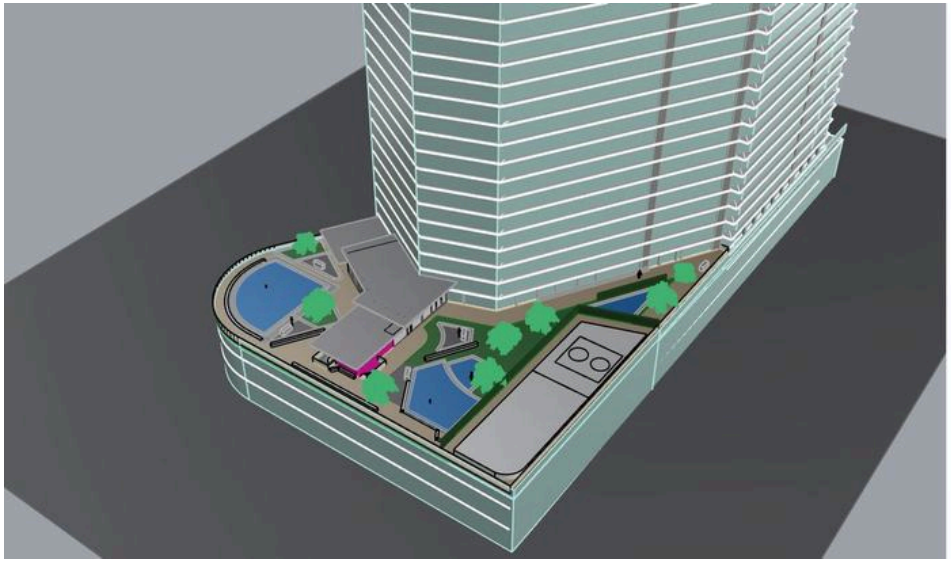
DESIGN NARRATIVE | THREE PRIVACY ZONES

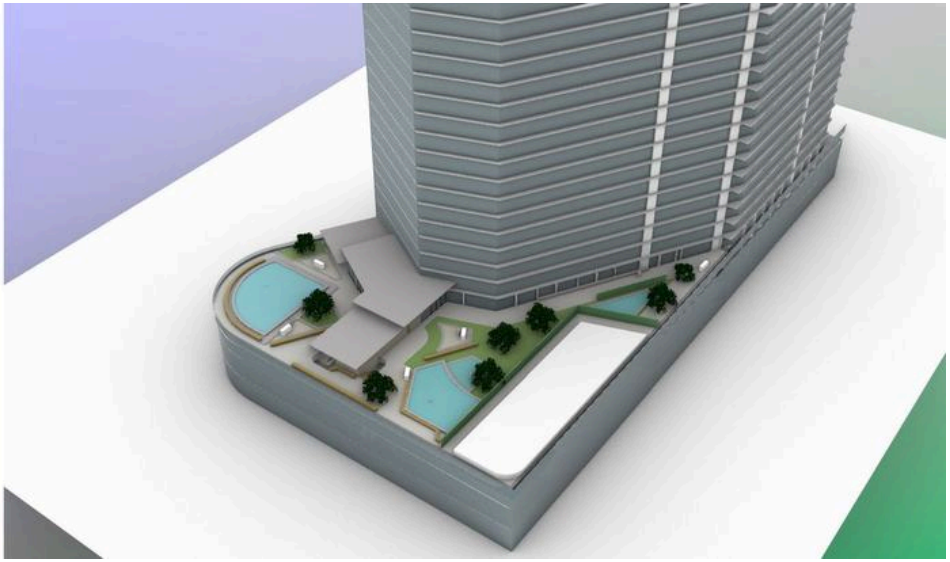
THE DECK



DRAWING | HARDSCAPE PLAN, THE DELIVERABLE

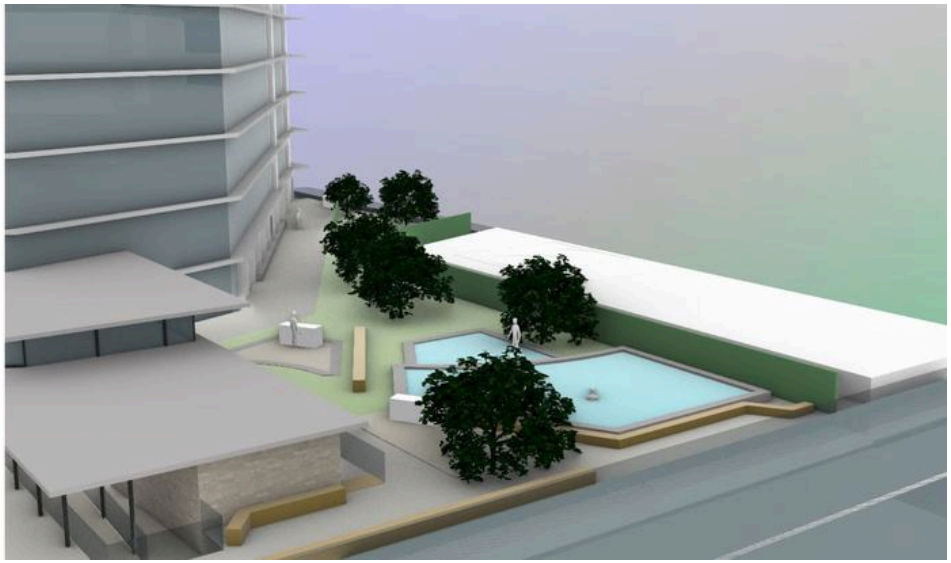
THE DECK





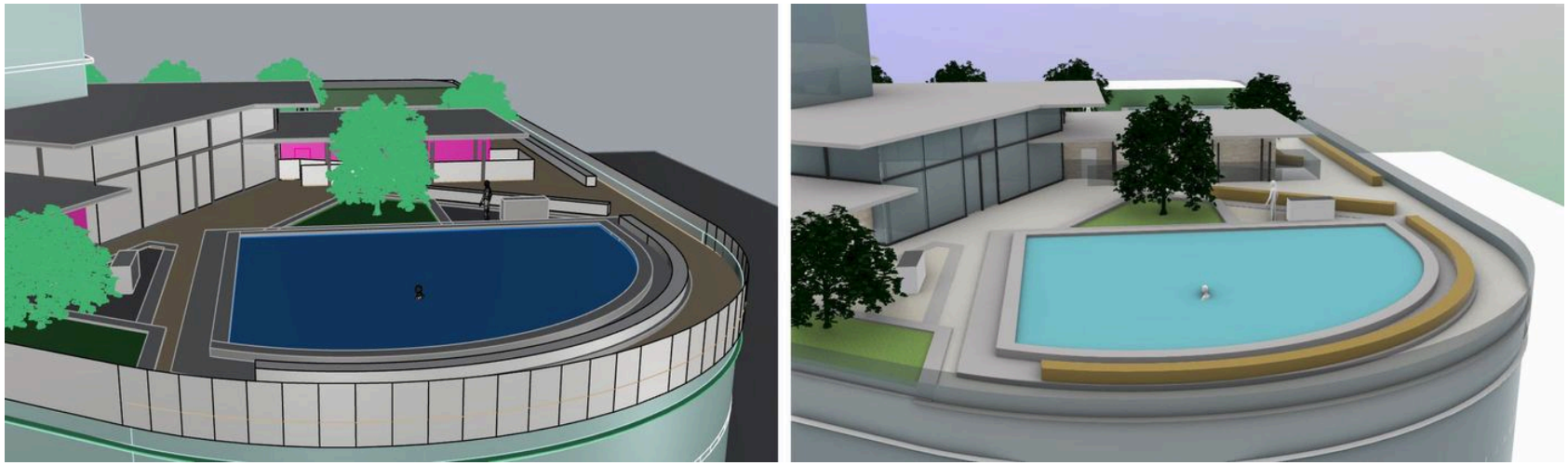
RHINO MODEL AND FINISHED RENDER | AERIAL



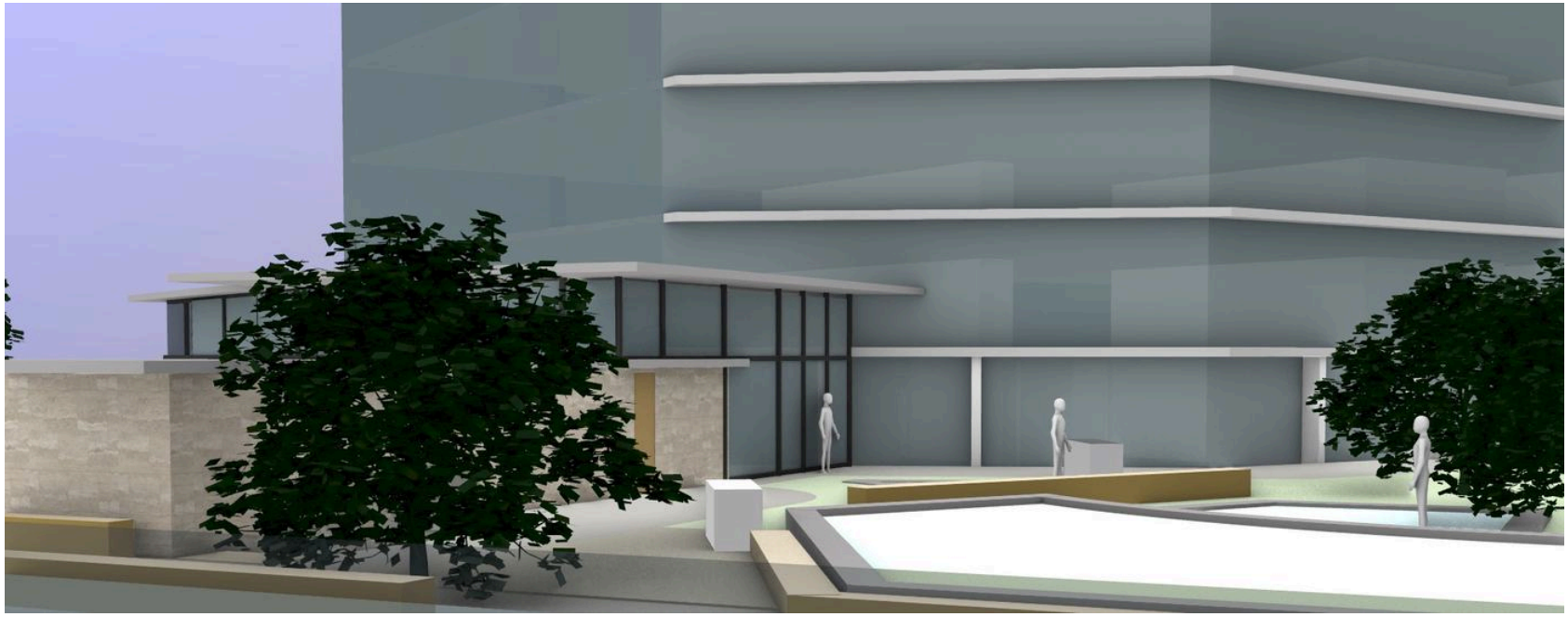


RHINO MODEL AND FINISHED RENDER | ZONE 2

THE DECK



RHINO MODEL AND FINISHED RENDER | ZONE 3



RHINO MODEL | PAVILION TERRACE

THE DECK



RENDER | PAVILION TERRACE (AI-ASSISTED)



03

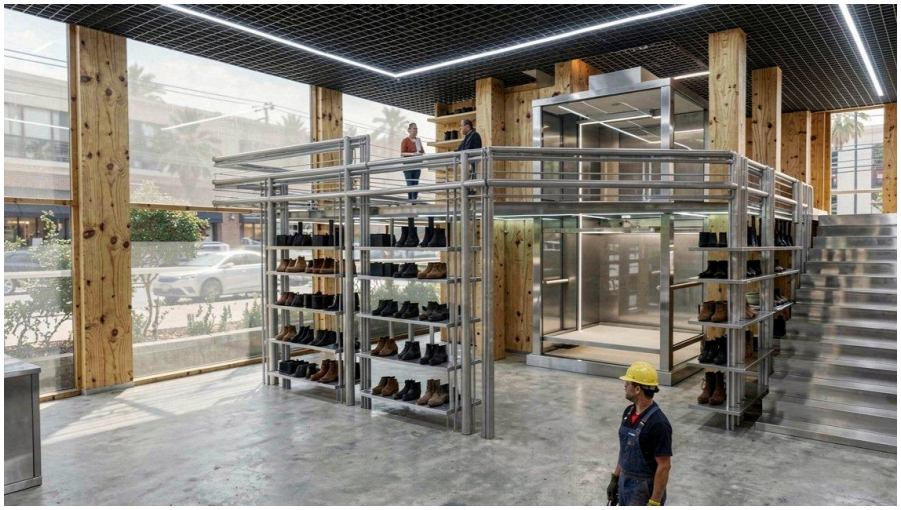
PROTOTYPE

SP26, PROTOTYPE, 3GB STUDIO

A generative pipeline for retail architecture and product design, built around a footwear brand's flagship storefront. Architectural iterations were produced through a custom Flux/ComfyUI pipeline, while the product line — a technical boot shown here in cutaway and colorway studies — was developed in parallel through a Gemini-based image pipeline. The two threads feed each other: the building's industrial material palette of raw wood, steel, and concrete is carried directly into the product's own construction and display.



CUTAWAY SECTION ELEVATION OF THE FULL BUILDING



INTERIOR RETAIL FLOOR WITH STEEL AND TIMBER SHELVING DISPLAY



INTERIOR COUNTER SCENE WITH STAFF AND CUSTOMER



TECHNICAL BOOT PRODUCT RENDER, STUDIO SHOT



BOOT COLORWAY COMPARISON SHEET, FOUR VARIANTS



TECHNICAL CUTAWAY OF THE BOOT'S INTERNAL CONSTRUCTION



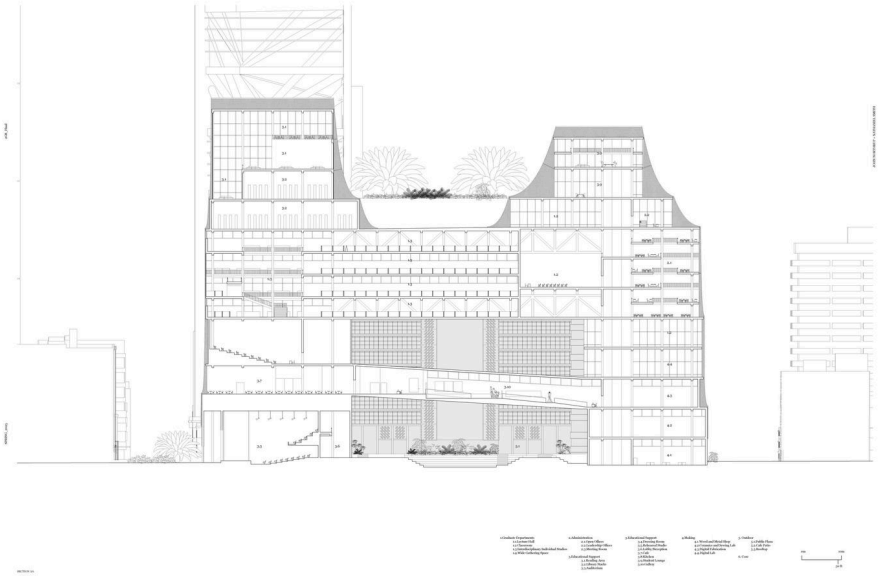
04

DESIGN SCHOOL IN MEXICO CITY

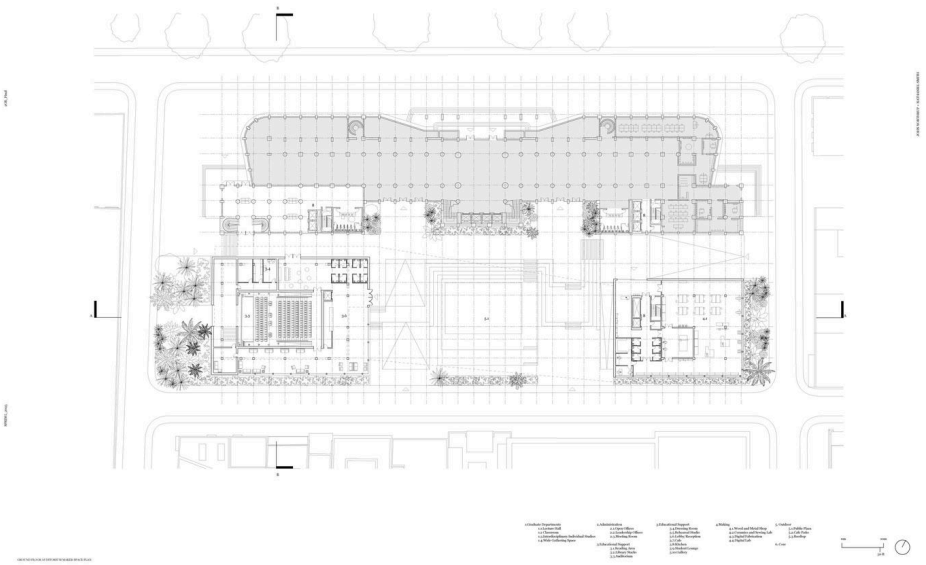
SP25, 2GB STUDIO, WITH NATE SMITH
ADVISOR · ZEINA KOREITEM

An addition to the IMSS headquarters, Mexico's social security bureau, in Mexico City: a design school that plugs into the existing office building without major changes to its structure. On our studio trip we read the city as a palimpsest, built in layers, with new buildings and materials laid directly against the ones before them. The proposal continues that reading. Studios, classrooms, a gallery, a café and a library stack as blocks on one another and on the IMSS building, a public plaza opens at street level, and a mesh facade drapes over the whole: one more layer of the city.

*Made with Nate Smith



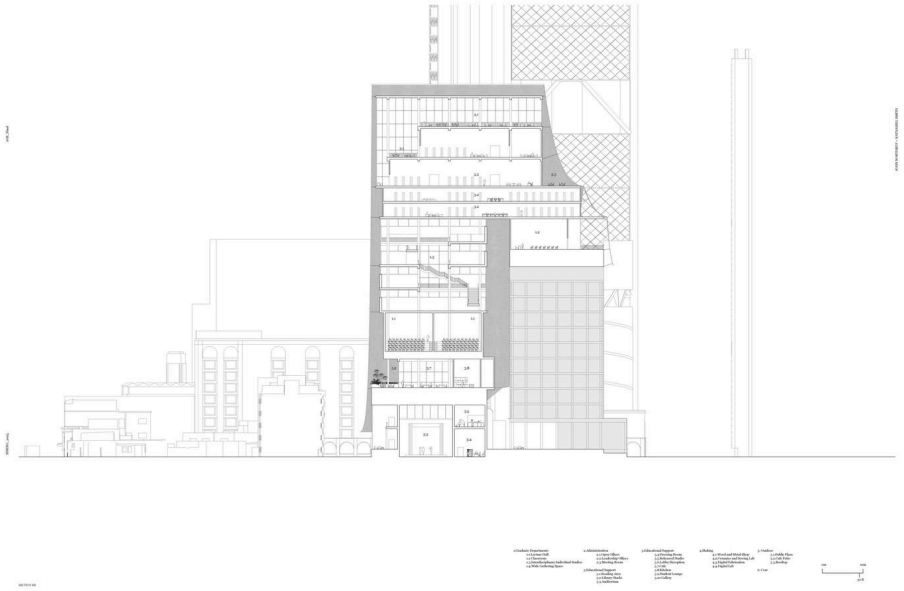
DRAWING | SECTION AA



DRAWING | GROUND FLOOR PLAN AND PUBLIC PLAZA



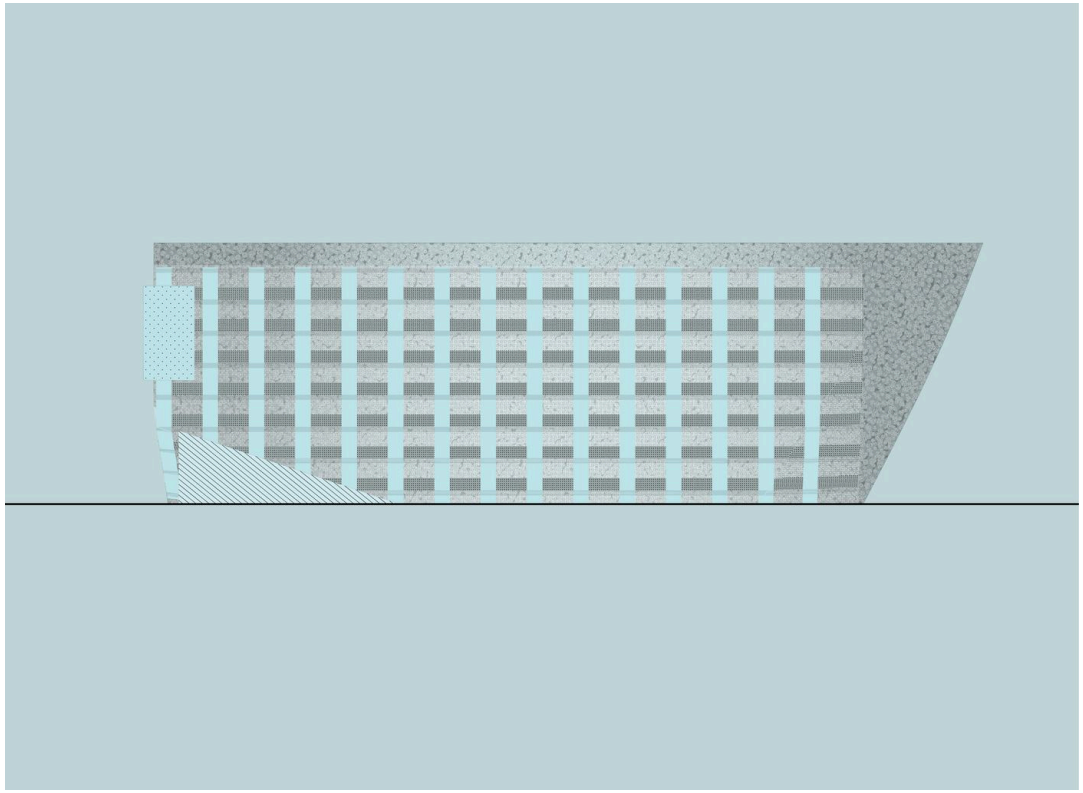
MODEL | BRICK-CLAD BLOCKS AND TRUSS BRIDGE



DRAWING | SECTION BB



MODEL | STUDY MASSING IN THE CITY SITE MODEL



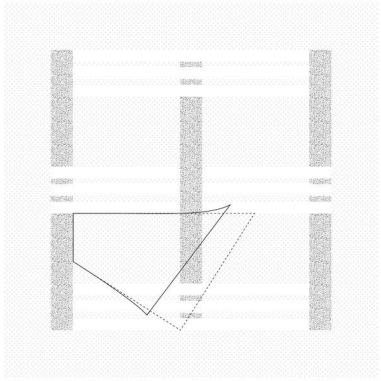
05

LITTLE TOKYO THEATER

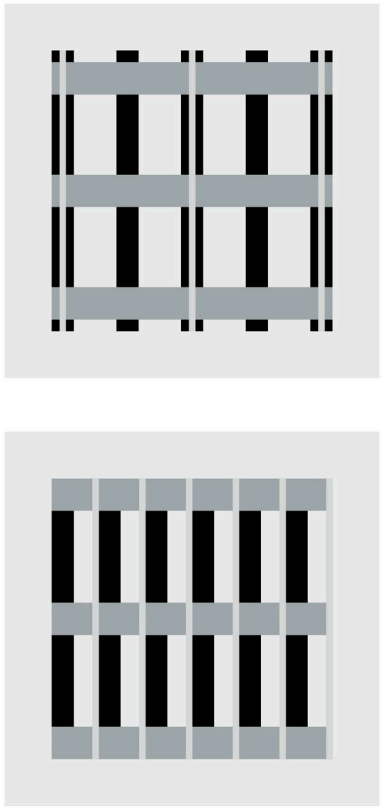
FA23, DS 1100, 1GA STUDIO
ADVISOR · MATTHEW AU

This first semester studio project stems from a study of grid systems, materiality, galleries, and theatres. The initial measurement grids were carefully constructed, using the limitations of the tape itself and the gap in named measurements vs. actual measurements. The grids operate in a way that looks at material falsities, as well as transparency. Through these measurement systems, they are then expanded into large boards. These boards are then used to produce a massing model. Ultimately realized as a performing arts theater in the Little Tokyo District, the Measurement Grid is transformed into a space that offers a place for community to gather in an unconventional theater setting where actors and patrons are active participants in the shows on display.

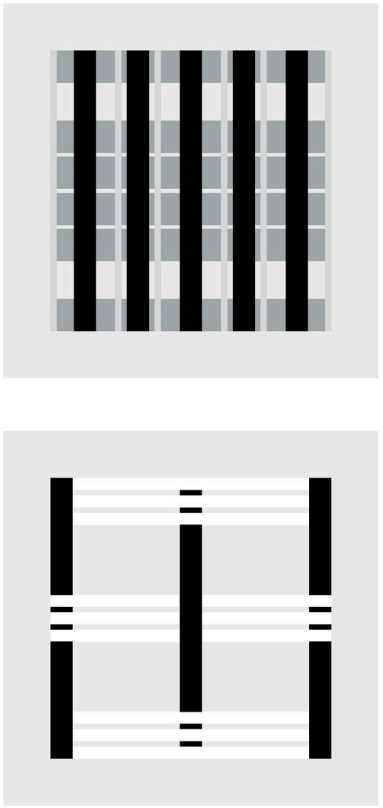
GRID STUDIES



MEASUREMENT GRID DIAGRAM WITH TETRAHEDRON

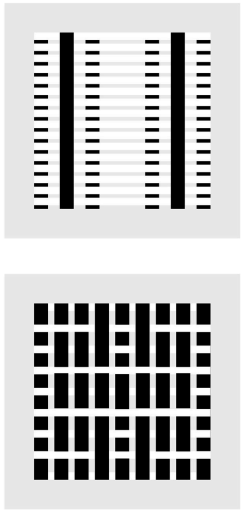


DIGITAL GRID FIGURE STUDY 1 AND 2

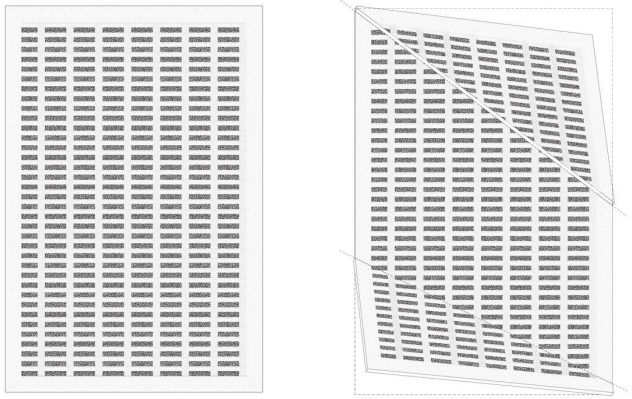


DIGITAL GRID FIGURE STUDY 3 AND 4

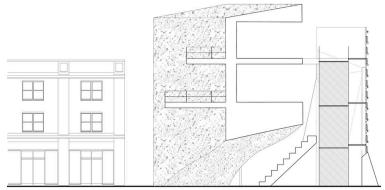
GRID STUDIES



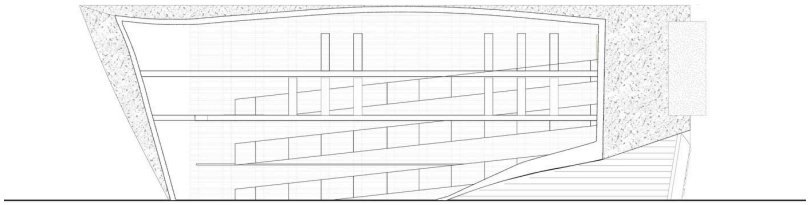
DIGITAL GRID FIGURE STUDY 5 AND 6



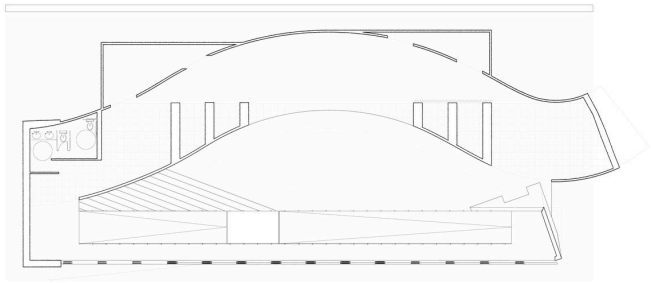
DIGITAL GRID STUDY, PRINTED BOARD FLAT AND BENT



GRID STUDY DRAWING, SHORT SECTION

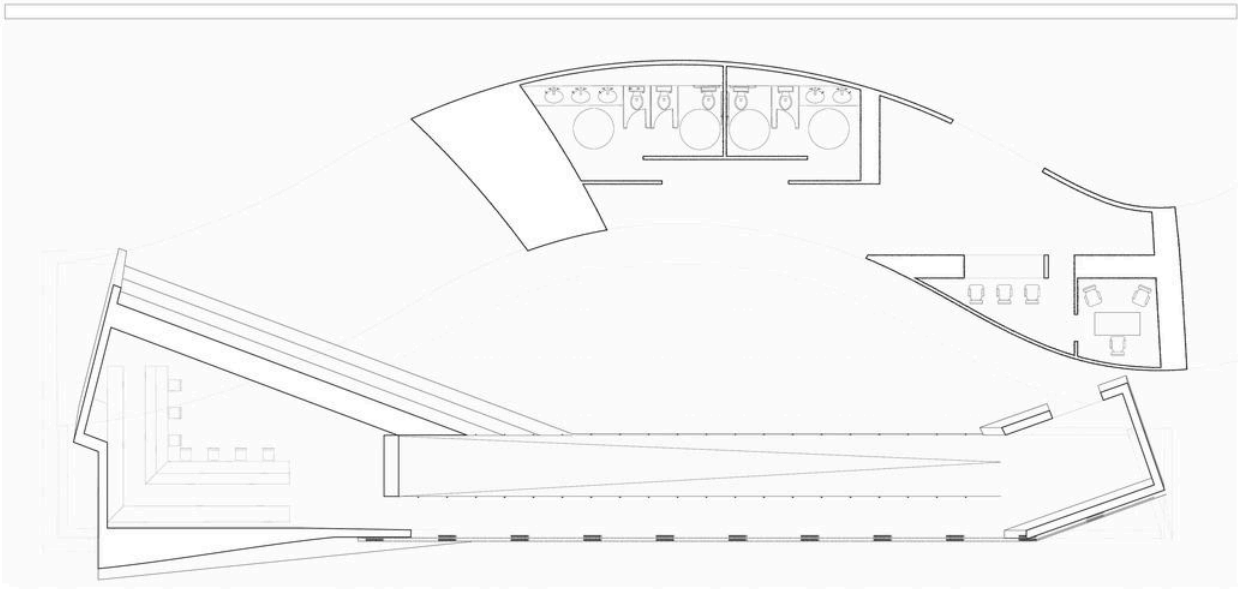


GRID STUDY DRAWING, LONG SECTION



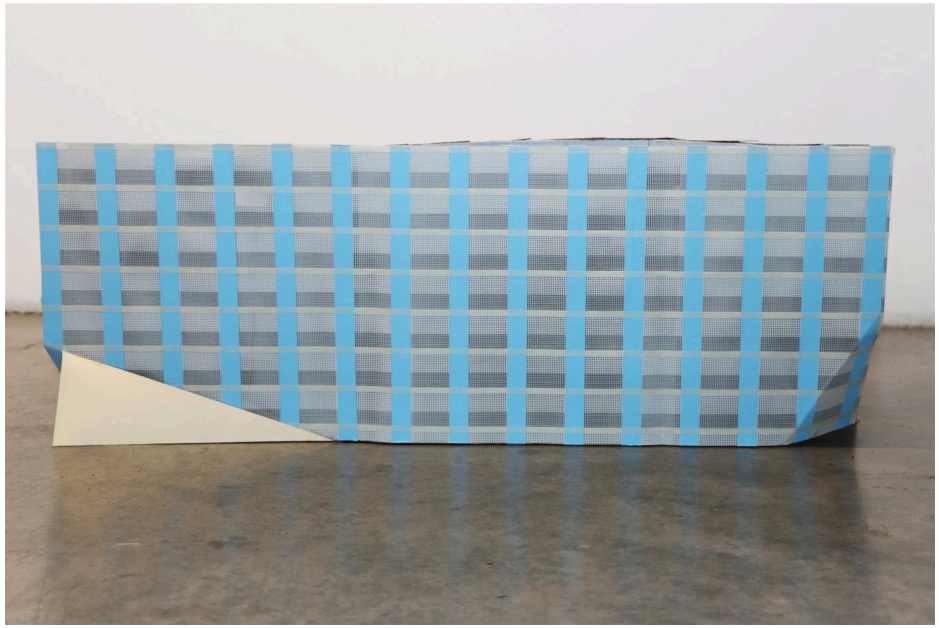
GRID STUDY DRAWING, SECTION

GRID STUDIES

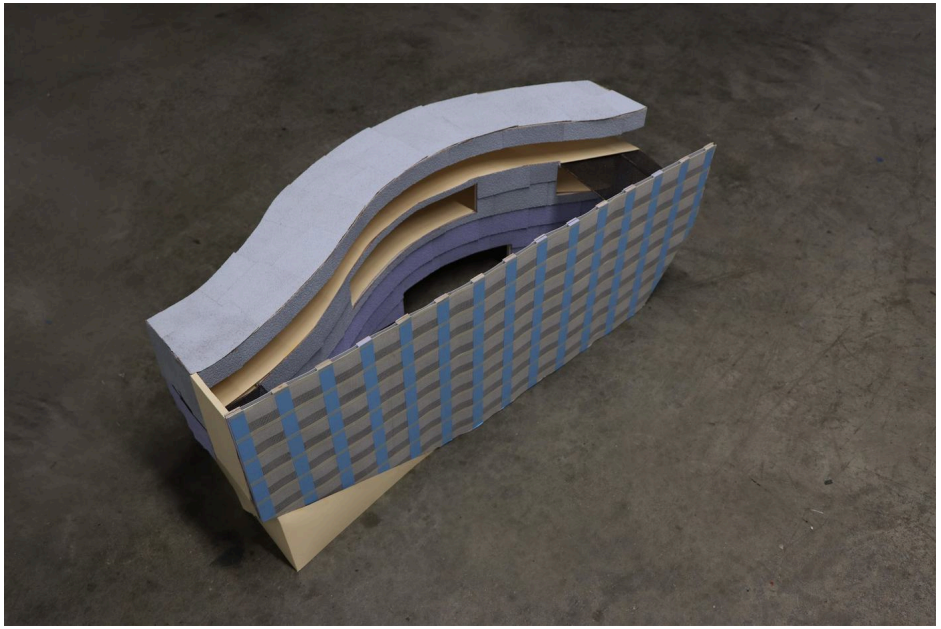


GRID STUDY DRAWING, PLAN

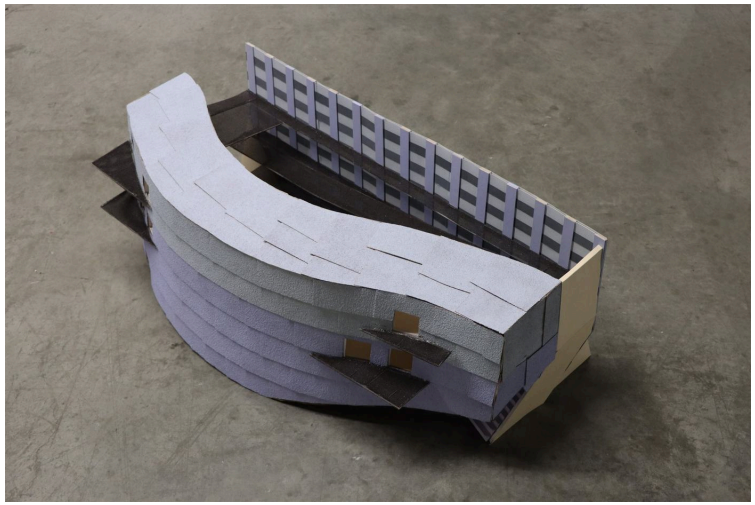
FINAL MODEL



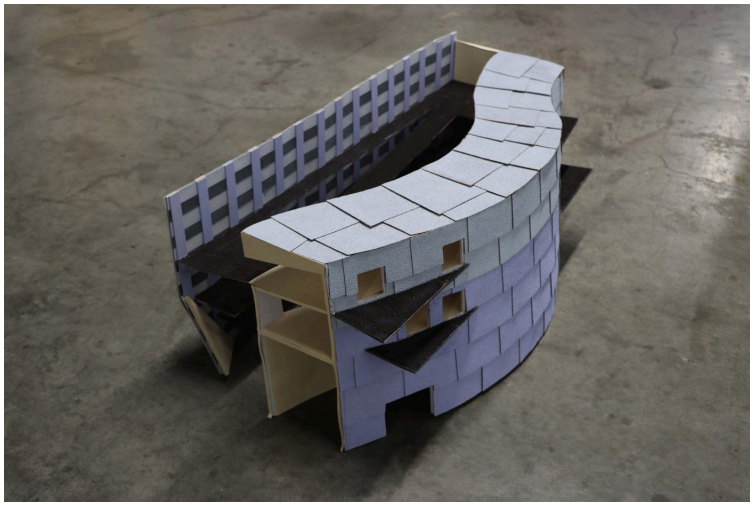
FINAL MASSING MODEL, PRIMARY VIEW



FINAL MASSING MODEL, VIEW 2



FINAL MASSING MODEL, VIEW 3

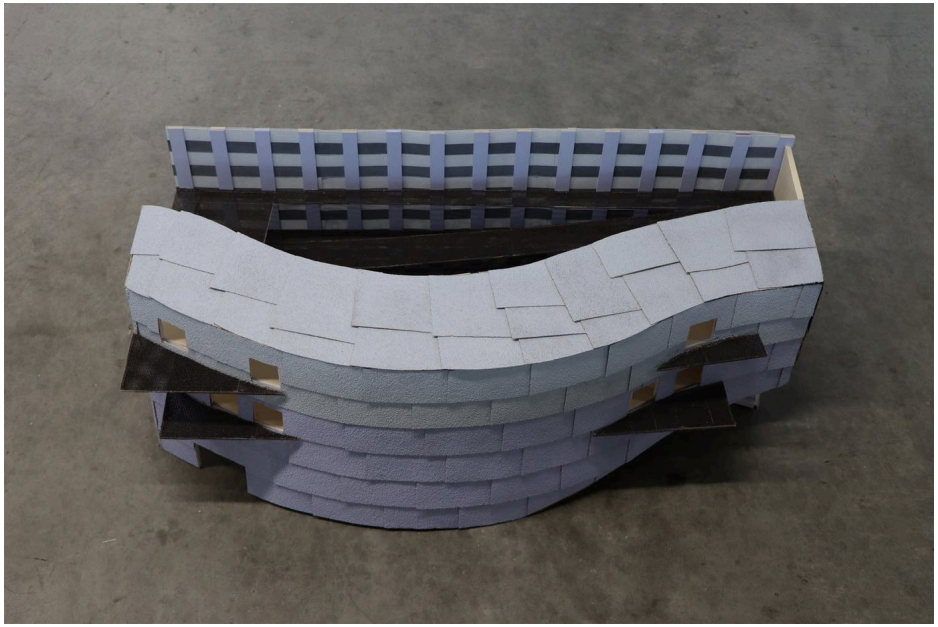


FINAL MASSING MODEL, VIEW 4

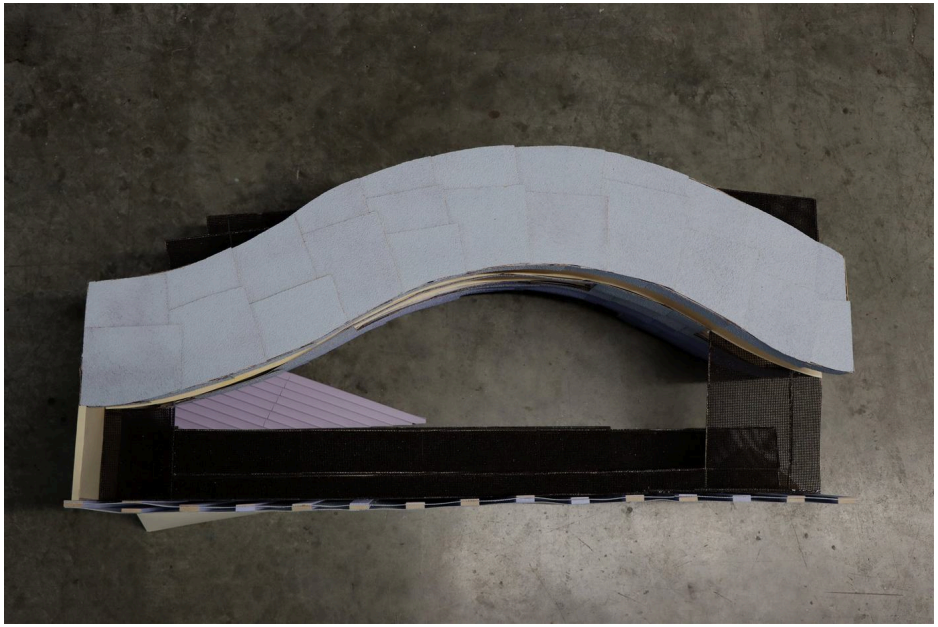


FINAL MASSING MODEL, VIEW 5

FINAL MODEL

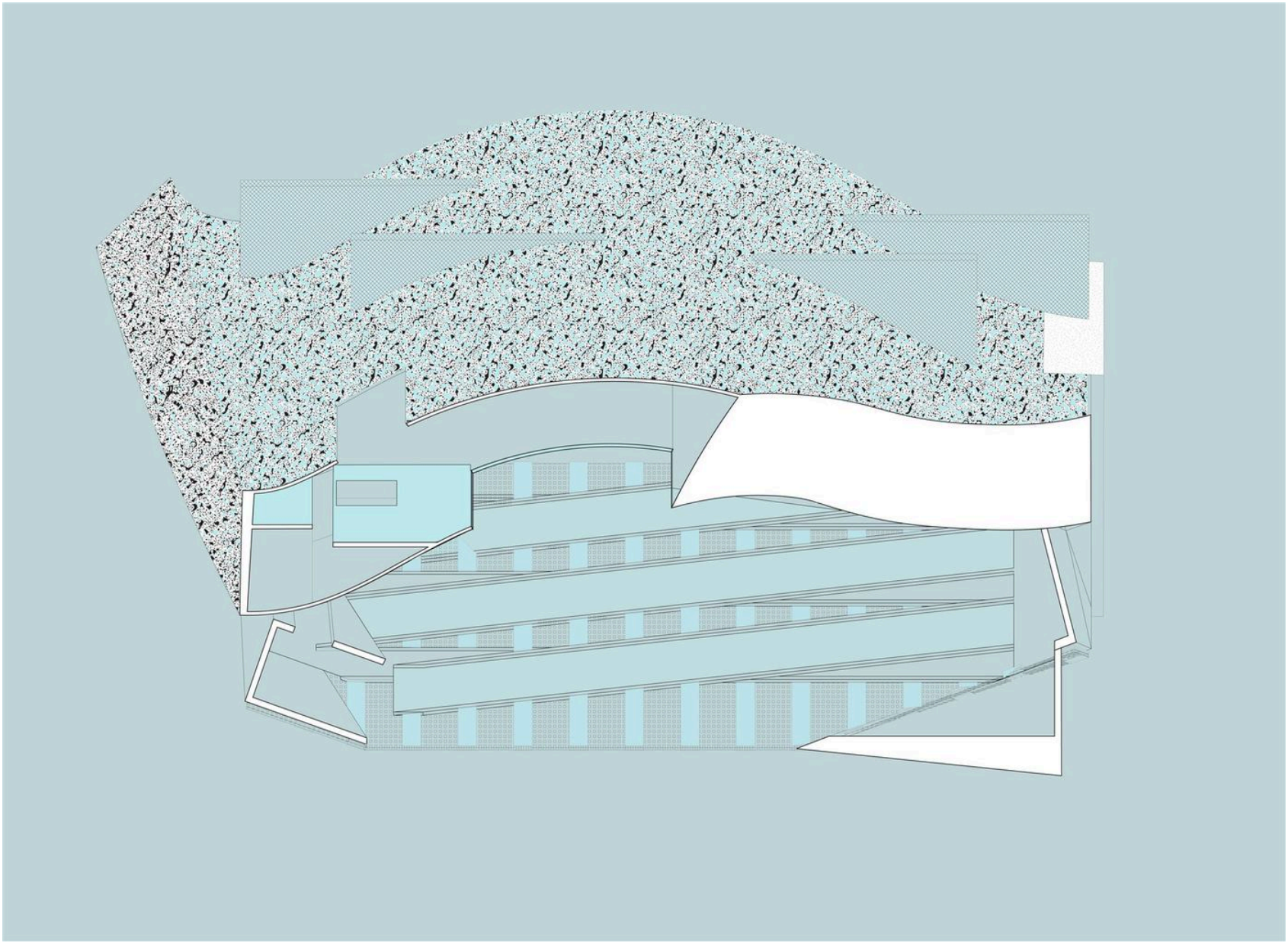


FINAL MASSING MODEL, ELEVATION VIEW

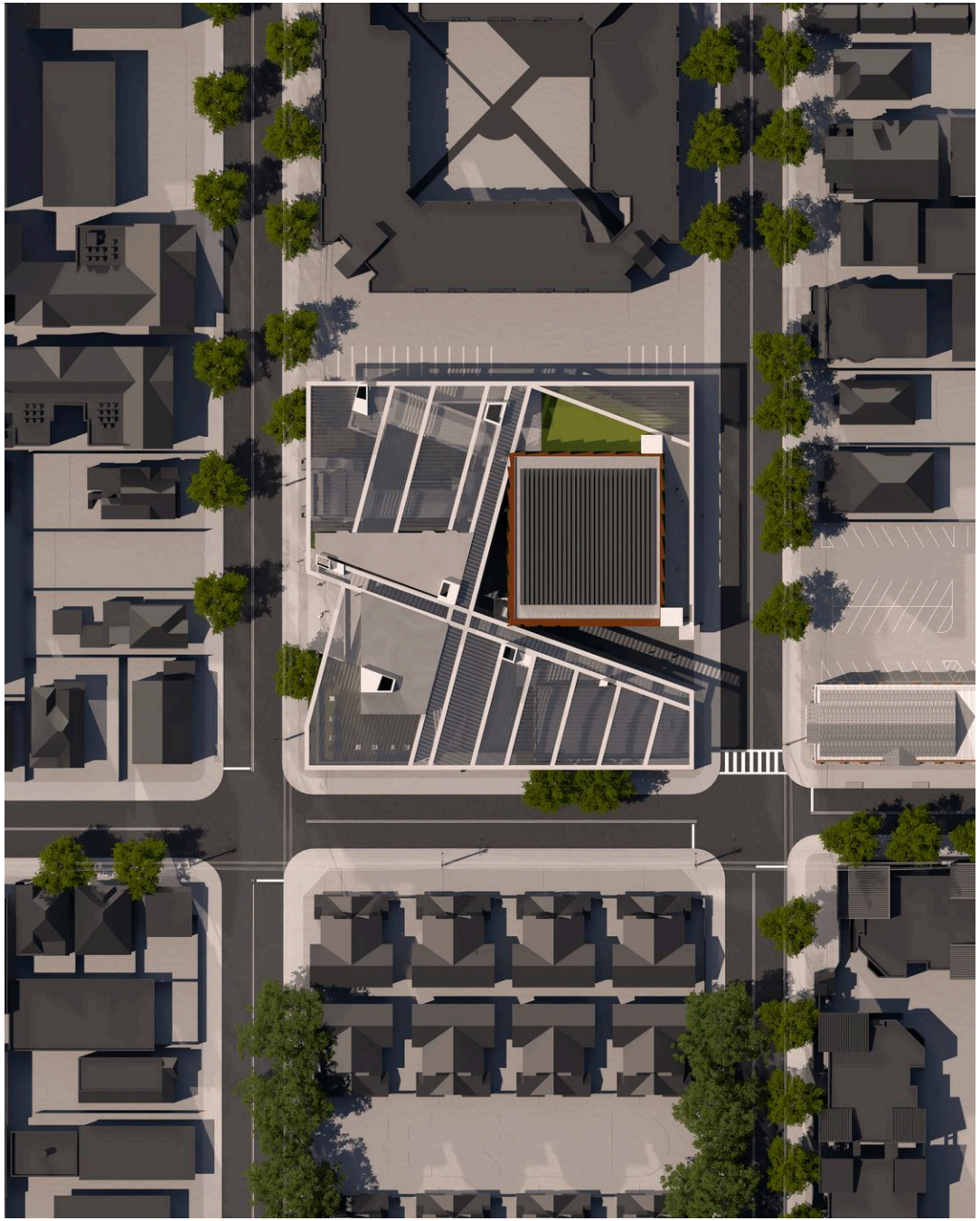


FINAL MASSING MODEL, VIEW 7

FINAL MODEL



FINAL MODEL COMPOSITE SECTION DRAWING



06

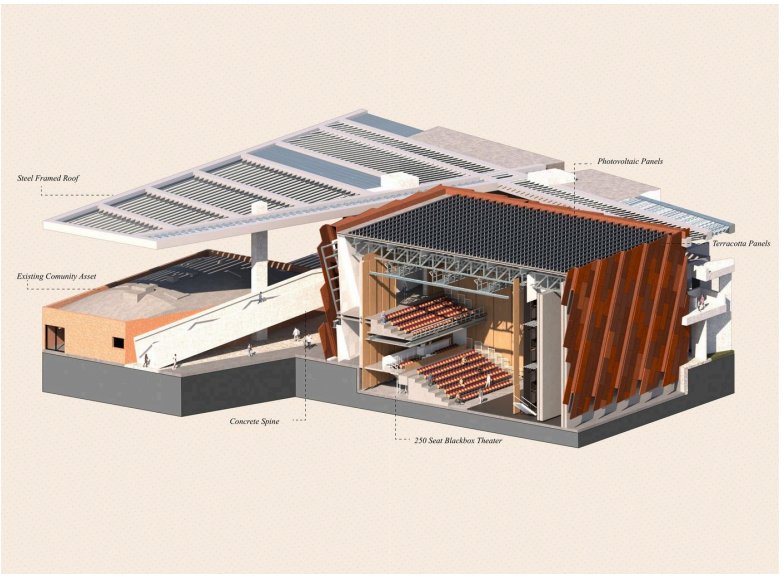
LA DANCE SHED

FA24, 2GA STUDIO, WITH MILAN SLEDGE
ADVISOR · MARCELO SPINA

This project for the Los Angeles Dance Project (LADP) seeks to bring the community of Pico-Union together on a campus that serves and connects the community. Located directly across from the Pico-Union Project, the site was surgically opened up in order to connect via sightlines and circulation to this important community asset. A unique goal among this project was to preserve many of the existing structures on the site and repurposing the materials from buildings that were to be removed. With a raised circulatory system, visitors are encouraged to meander throughout the site, engaging with it from within the in-between spaces. The roof serves as not only shade for warm summer days, but also acts as an organizing device for the discrete elements of the site, bringing them together in a collective, just as is intended with the surrounding community.



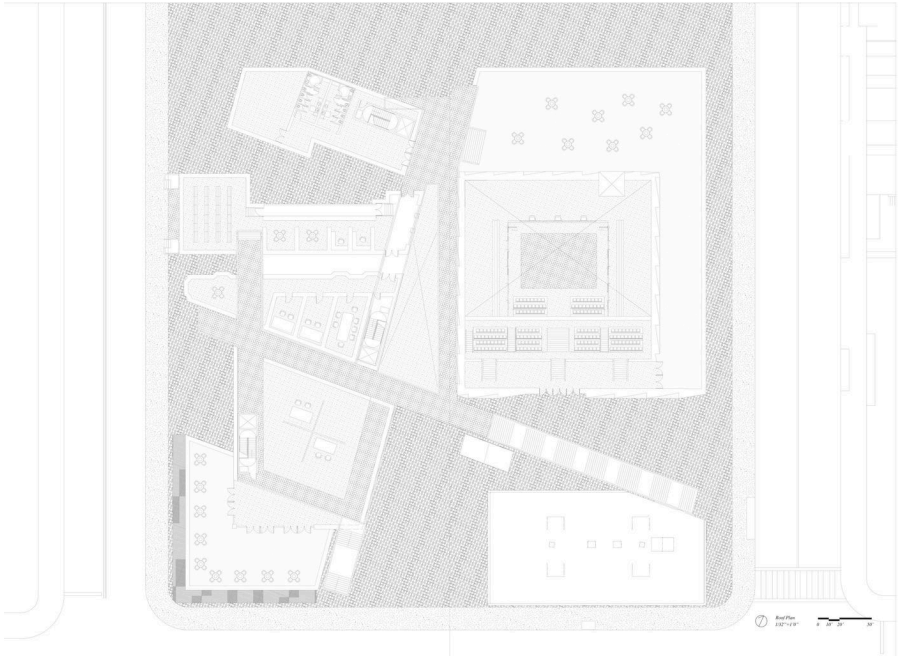
RENDER | FROM THE PICO-UNION SIDE OF THE SITE



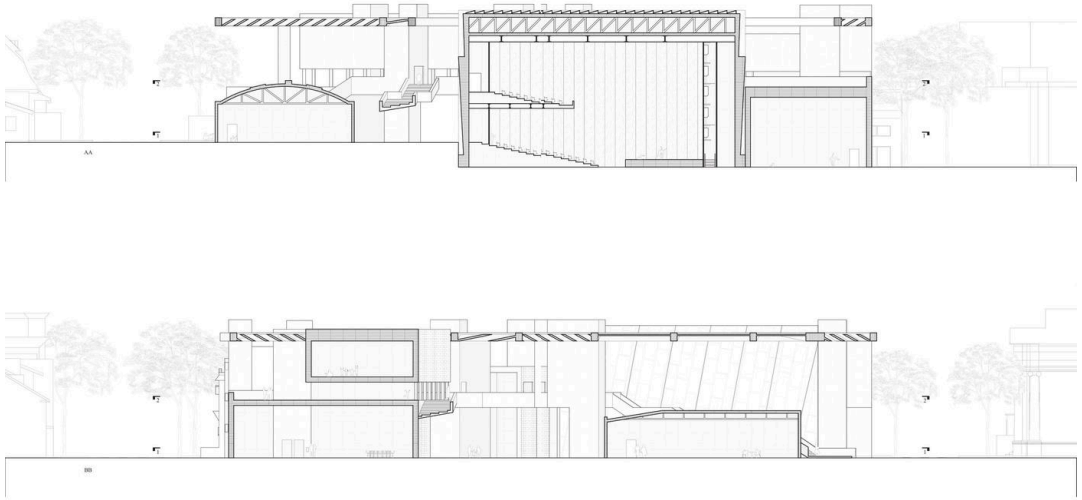
AXONOMETRIC | STEEL-FRAMED ROOF, CONCRETE SPINE AND BLACK BOX THEATER



PLAN | LEVEL 1



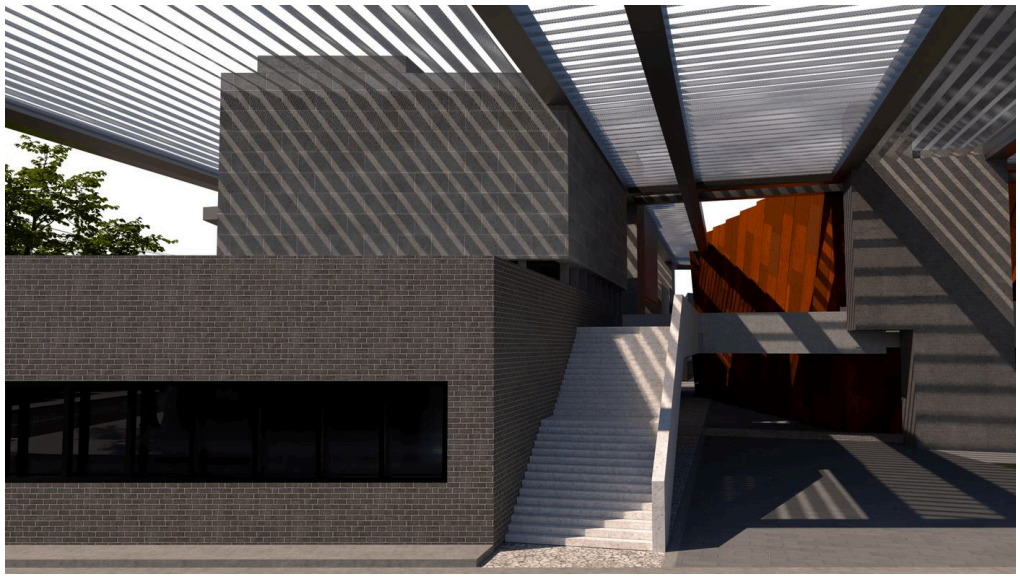
PLAN | LEVEL 2



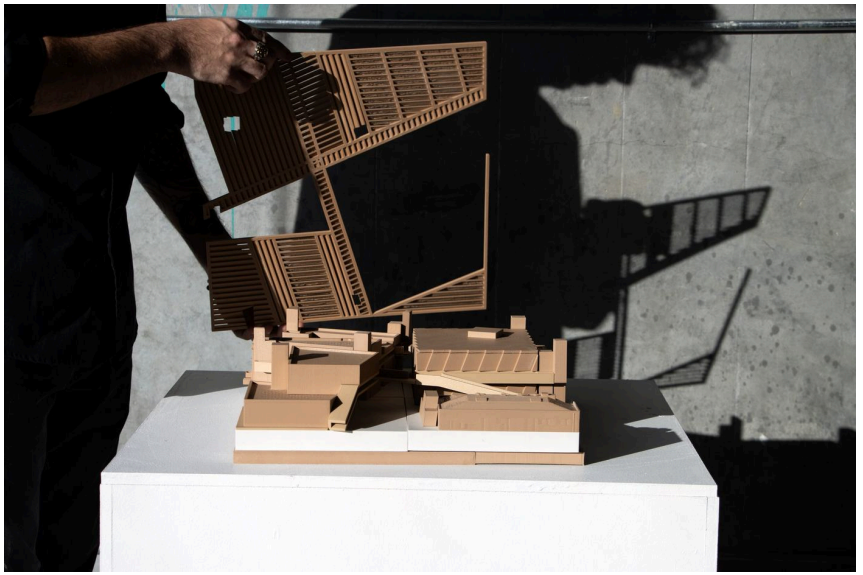
SECTIONS | THROUGH THE THEATER AND SPINE



ELEVATIONS | STREET FRONTAGES, TERRACOTTA PANEL FACADE



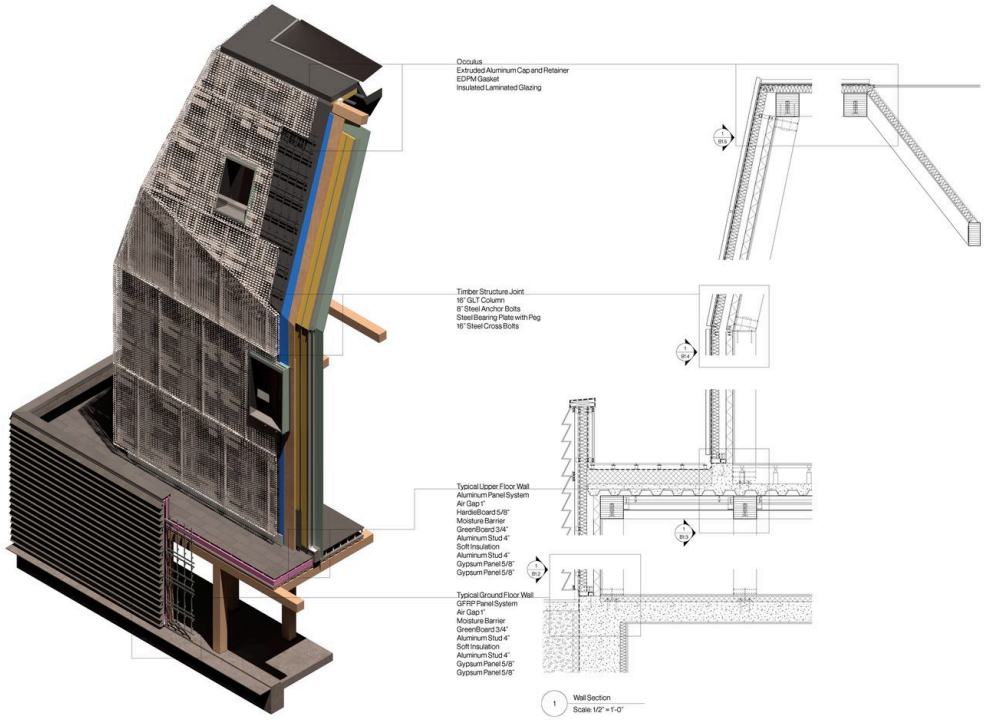
RENDER | TOWARD THE CAMPUS CAFE



MODEL | THE ROOF LIFTED OFF THE CAMPUS BELOW



MODEL | SETTING THE ROOF PLATES OVER THE SITE

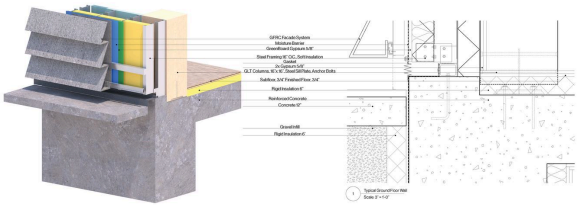


07

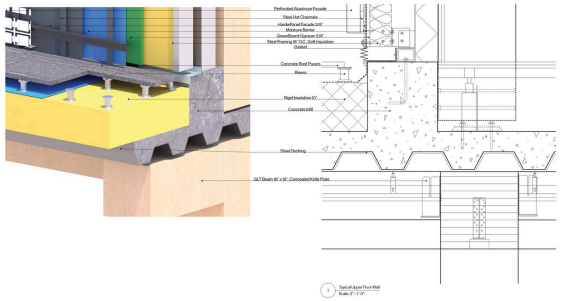
DESIGN DOCUMENTS

DESIGN DOCUMENTS
ADVISOR · HERWIG BAUMGARTNER

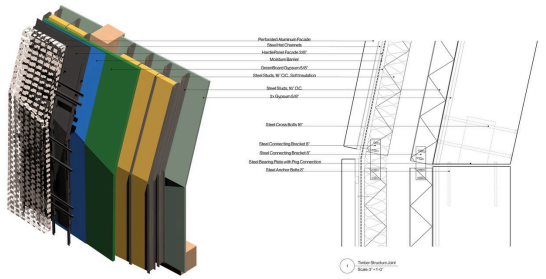
This course studies the implementation of design, technology, and an understanding of building systems. Students are tasked with preparing Design Document drawings from a previous year's studio project and are asked to bring it to life with real-world materials, budgets, and environmental factors.



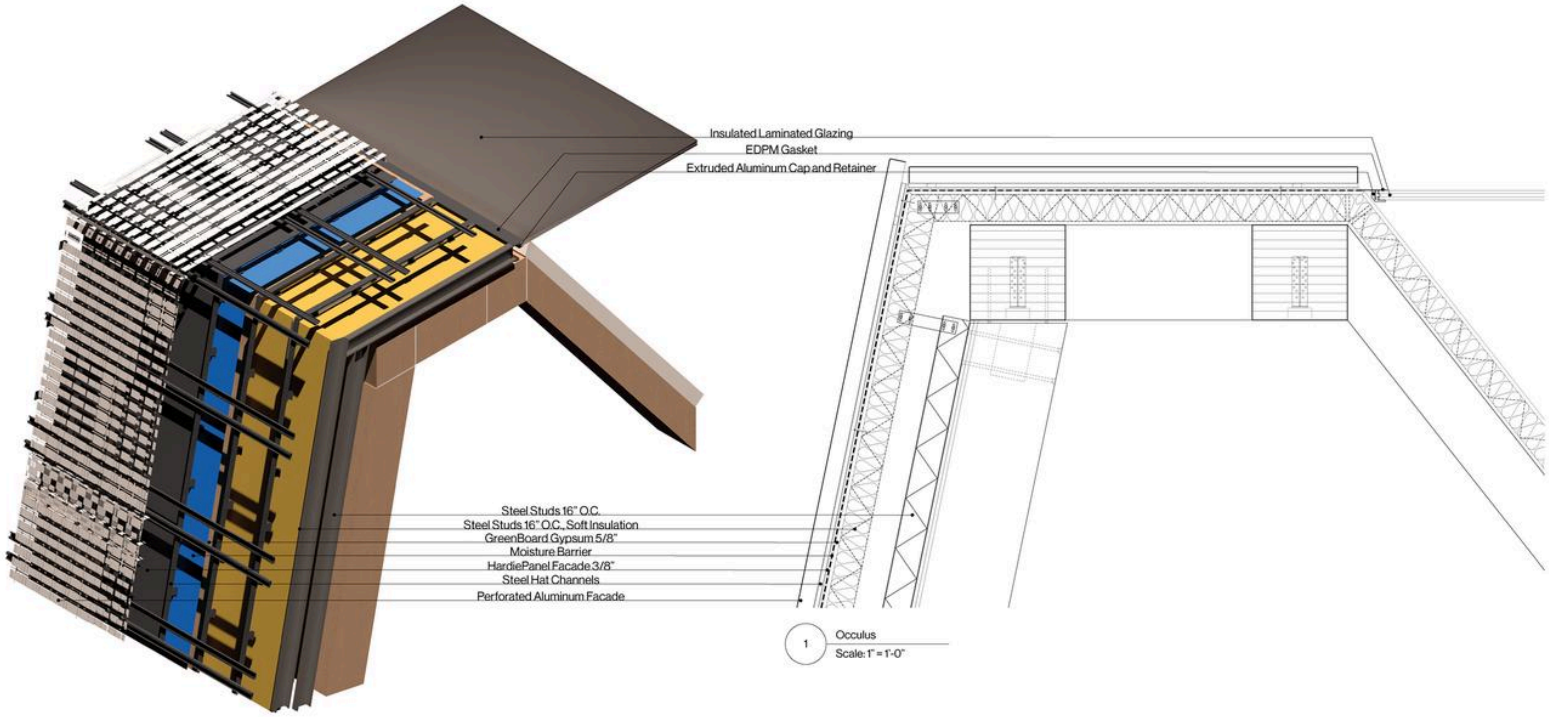
TYPICAL GROUND FLOOR WALL — PEEL-BACK RENDER AND DRAWI...



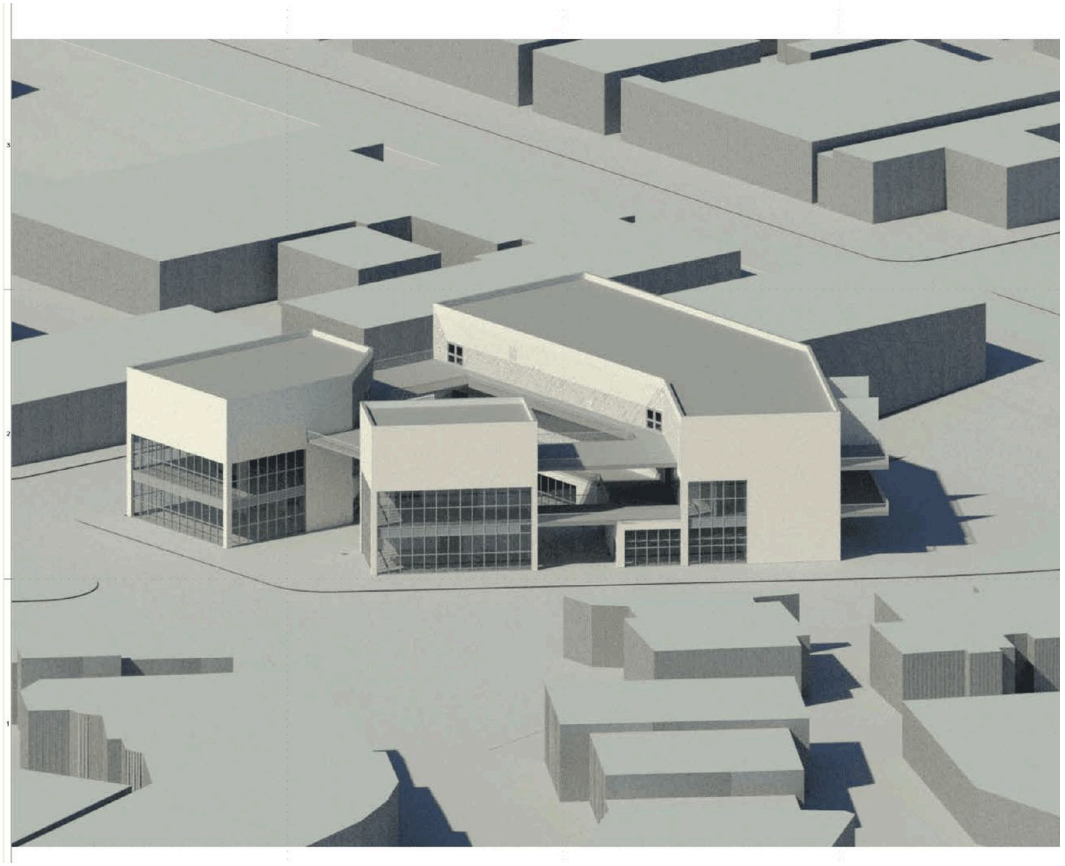
TYPICAL UPPER FLOOR WALL — PEEL-BACK RENDER AND DRAWING



TIMBER STRUCTURE JOINT — PEEL-BACK RENDER AND DRAWING



OCULUS SKYLIGHT — PEEL-BACK RENDER AND DRAWING



08

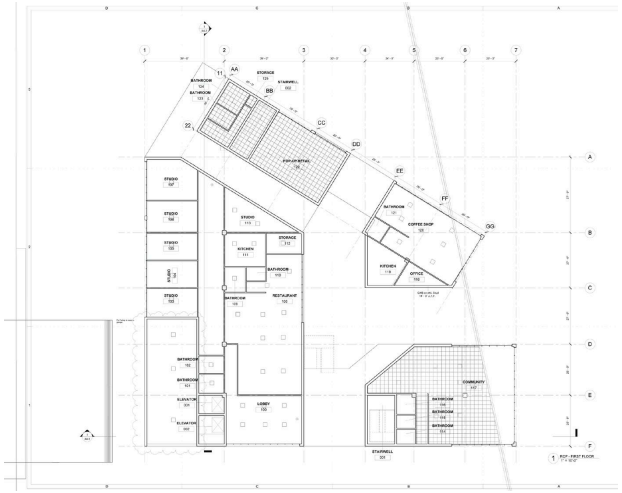
ADVANCED PROJECT DELIVERY

ADVANCED PROJECT DELIVERY
ADVISOR · PAVEL GETOV AND KARENZA HARRIS

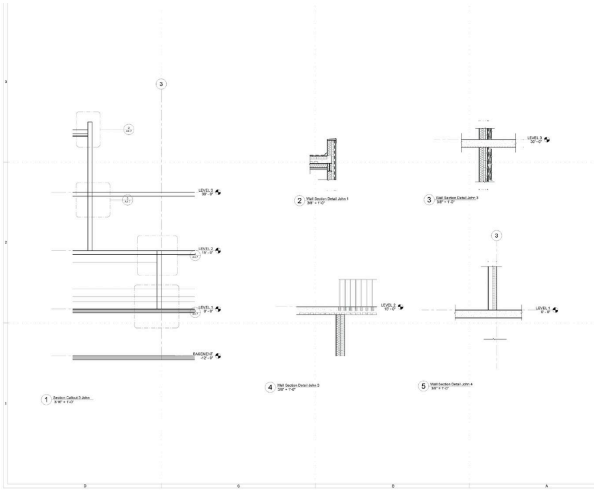
A Construction Documents course focused on taking a design through to buildable detail — resolving material assemblies, structural connections, and building systems into a coordinated drawing set. Site plans, reflected ceiling plans, and wall section sheets were developed to real-world standards of dimension, specification, and constructability.



SITE PLAN SHEET



REFLECTED CEILING PLAN, FIRST FLOOR



DETAILS AND WALL SECTION SHEET



09

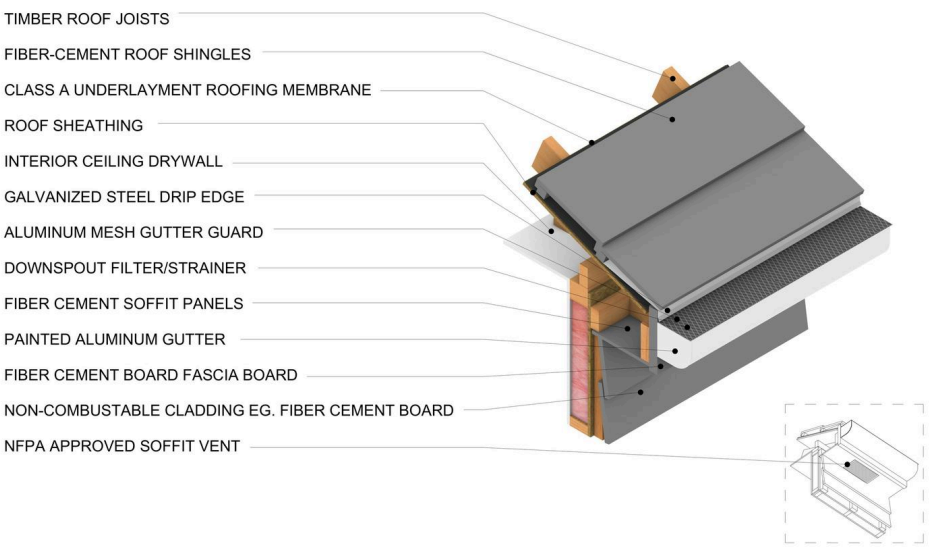
RESILIENT FUTURES

2025, SCI-ARC RESILIENT FUTURES TASK FORCE
ROLE · DRAWINGS, DESIGN & FABRICATION

SCI-Arc formed the Resilient Futures Task Force after the January 2025 firestorms that burned through Altadena and the Pacific Palisades. One thing the fires left behind was suspicion of timber: people rebuilding wanted to know whether building in wood had made it worse.

These drawings and models were made for the task force's public work on that question, alongside its Demystifying Timber session. Nothing in them is exotic. They are the ordinary parts that make an ordinary timber house more likely to survive a fire, and the whole point is that you can see every one. Three conditions are drawn as annotated axonometrics and then built at full scale: the eaves and overhang, the wall, and the deck. I drew all three, and helped design and build the eaves and overhang.

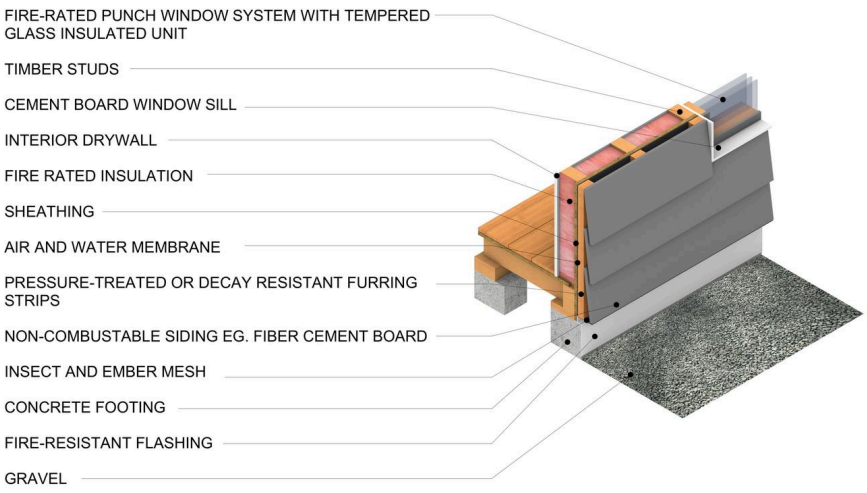
The specification is the argument. Fiber-cement shingles over a Class A underlayment membrane, non-combustible fiber-cement cladding, soffit and fascia, NFPA-approved soffit vents, an aluminum mesh gutter guard and a downspout strainer to keep embers out of the gutter, fire-rated insulation, and a fire-rated window with tempered insulated glass. Built at full size, the vent screen and the gutter guard stop being line weights and become things you can stand in front of.



DRAWING | EAVES AND OVERHANG, LAYER BY LAYER



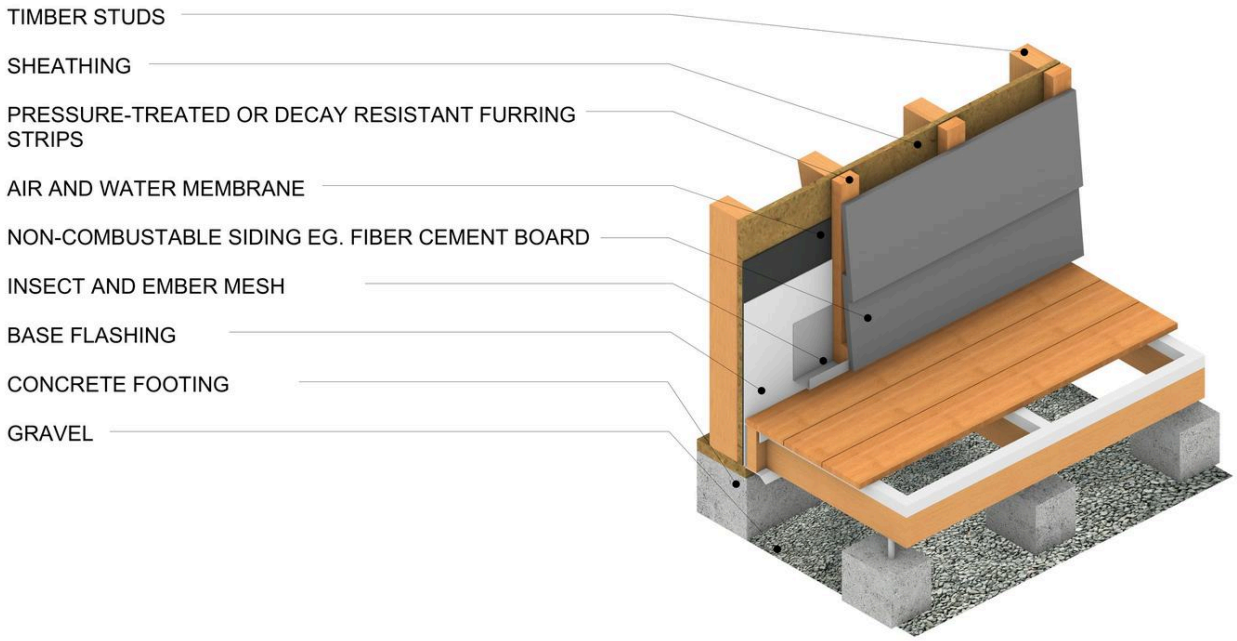
MODEL | THE VENTED SOFFIT AND GUARDED GUTTER, FROM BELOW



DRAWING | WALL, WITH THE FIRE-RATED WINDOW AND SILL



MODEL | THE TIMBER FRAME BEHIND THE CLADDING



DRAWING | DECK, FURRING AND NON-COMBUSTIBLE SIDING



MODEL | THE ASSEMBLY AT FULL SCALE



10

DETAILS, DETAILS

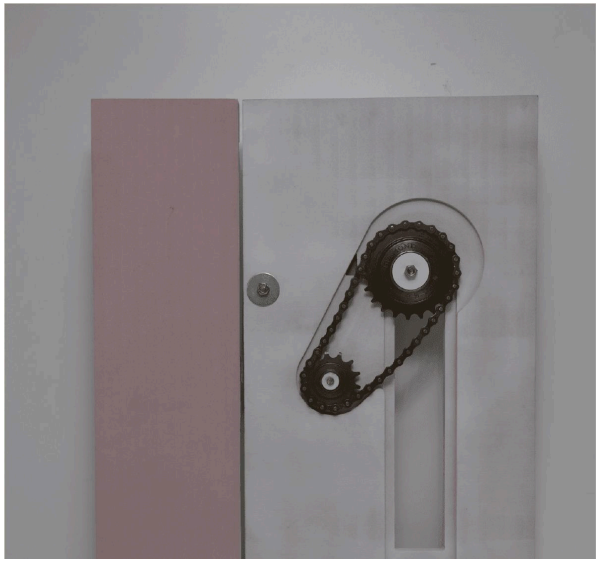
F25, DETAILS, DETAILS
ADVISOR · DWAYNE OYLER

A doorlock mechanism inspired by Frank Lloyd Wright's detail work, specifically the door mechanisms in the Hollyhock House.

*All work made with partners Chia-Chen Li and Baizhen Yu



FOAM PROTOTYPING PROCESS



PROTOTYPE INTERIOR MECHANISM



FINAL DOOR MECHANISM, CLOSE-UP



11

ELF-O-MATIC-3000

FA22, MONSTER CITY STUDIOS
ROLE · TECHNICAL DESIGNER & PROJECT MANAGER

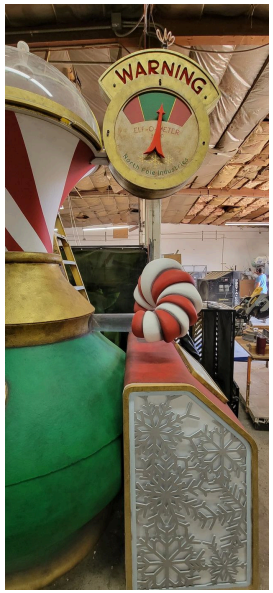
A holiday show piece built at Monster City Studios: a steam-powered candy machine with a neon marquee, a lit viewing window of tumbling balls, a STEAM grille, a pressure gauge and a candy-striped dome. Behind the theming is a working show, with lighting, moving parts and an operator panel to reset it between guests.



BUILT WORK | THE MACHINE RUNNING IN THE SHOP (VIDEO: JCNORTHRUP.COM/PROJECTS/E...



BUILT WORK | MARQUEE, GAUGES AND DOME



DETAIL | WARNING GAUGE



DETAIL | OPERATOR PANEL



DETAIL | SERVICE HATCH WITH LABELED CONTROL SWITCHES